

How Do Nonprofits Use Cash Windfalls? Evidence from \$5B in Unrestricted Donations*

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Abstract

I use \$5B of MacKenzie Scott’s unrestricted gifts as a natural experiment to test the hypothesis of a “nonprofit starvation cycle” caused by donation use-restrictions. Comparing 643 recipients to similar nonrecipients with difference-in-differences, I find that recipients attracted additional donations, spent the entire gift within two years, and disbursed 20% of new spending as grants to others and 3% as executive compensation. In present value, \$0.18 per dollar of the windfall was spent on executive pay. With a corporate-finance model of nonprofits maximizing charitable output subject to donation use-restrictions, the evidence casts doubt on the starvation cycle hypothesis.

Keywords: charitable giving, nonprofit operations, capital budgeting

JEL Codes: L31, G32, G35, H41

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Large gifts to U.S. 501(c)(3)’s (nonprofits) are typically given with restrictions on use, but unrestricted gifts increased from 30% to 42% of foundation grants to nonprofits from 2016-2022.¹ Unrestricted gifts, as their name suggests, offer recipient nonprofits full discretion over the use of funds across their portfolio of charitable projects and across their direct and indirect costs. This flexibility stands in contrast to traditional, project-based gifts, which earmark funds for direct costs or for particular projects within the nonprofit.

The use of such gifts can answer two questions about nonprofit operations. First, what are the returns to scale of nonprofit operations? If nonprofits seek to maximize charitable output, the concavity of production mediates normative conclusions about nonprofit market structure and competition (Lapointe et al., 2018). Second, do nonprofits face a “starvation cycle” that inhibits investment in productivity-increasing indirect costs and saving that unrestricted gifts are intended to prevent? Unrestricted gifts and “trust based philanthropy” have grown in popularity due to a belief that earmarks and spending requirements hamper the ability for nonprofits to allocate funds most effectively (Gregory and Fall, 2009). On the other hand, releasing such earmarks could lead to the redistribution of funds from external to internal stakeholders via agency frictions that the earmarks successfully prevent (Fama and Jensen, 1985). Confirming one of these stories has been challenging to date because donations to nonprofits largely carry restrictions.

In this paper, I show that under a model of a nonprofit agent maximizing charitable output and private benefits subject to constraints, changes in the allocation of spending at nonprofits following an exogenous wealth shock can answer both questions. I test the predictions of this model using a sample of philanthropist MacKenzie Scott’s unanticipated unrestricted gifts, which ranged from less than \$1M to \$50M, made to hundreds of nonprofits between 2020-2022. Scott’s gifts during this period were unrestricted, unanticipated, and large so the true effect on financial decision making can be easily identified from panel data. Large increases in “on-granting,” granting the received funds to other organizations, suggest decreasing returns to scale in charitable output. I find a lack of significant change in tests for starvation cycle constraints. In contrast, increases in executive compensation amounting to \$0.18 per dollar of the windfall when valued in perpetuity are consistent with agency frictions.

Data on nonprofit outcomes are derived from the population of tax forms filed by 501(c)(3)’s and posted online by the IRS. The Form 990 “Return of Organization Exempt From Income

¹Foundations, corporations, and individuals donate approximately \$600B to nonprofits annually (Indiana University Lilly Family School of Philanthropy, 2026). Nonprofits also receive at least \$300B in grants annually from governments (Clerkin et al., 2025). By way of the regulation (“Uniform Guidance” 2 C.F.R. Part 200), all U.S. federal government grants are restricted use (U.S. Office of Management and Budget, 2024).

Tax” includes comprehensive financial information at the nonprofit level such as granular breakdowns of revenue sources (e.g., contributions, program services, investment income) and expenses including grants to other organizations, employee compensation, and indirect costs such as office and travel expenses for nonprofits with gross receipts exceeding \$200K annually. I limit the sample to recipients reported by Scott’s foundation, Yield Giving, who received a single gift between 2020 and 2022. The resulting sample of 643 gifts totaled over \$5B and were on average \$8.4M in size. Relative to recipients, the gifts were massive, averaging 87% of the revenue received by recipients in 2019. The nonprofits, in their size, were representative of organizations that provide the median dollar of U.S. charitable spending. The median recipient had an annual budget of \$7.9M, and 91% of 2019 spending by 501(c)(3) Form 990 filers came from nonprofits above this size.

In order to identify the average effect of the gift, I match each recipient to four financially comparable nonprofits then use difference-in-differences to compare recipients to the matched population. Nearest neighbor matching on expenses, donations, dollars allocated to grants, and number of employees yields a sample of recipients that evolved in parallel to the matched control between 2012 and receipt of the gift. With the assumption of parallel trends, difference-in-differences identifies the average effect of the gift on recipient nonprofits. This identification relies on the assumption that nothing simultaneous to receiving a gift from Ms. Scott differentially affected treated nonprofits versus untreated nonprofits conditional on sector-by-year fixed effects. The variation in gift years between 2020-2022 and the unexpected nature of the gifts bolster the credibility of this identification strategy.

I then present a model formalizing the assumptions under which the allocations of spending to grants, indirect costs, savings, and compensation after a wealth shock can yield insights about the nonprofit production function. The model posits a nonprofit agent allocating wealth and donations optimally to inputs in the production of charity. Proposition 1 shows that the change in the grant spending share after a wealth shock reveals whether nonprofit production is increasing, decreasing, or constant returns to scale. Proposition 2 shows that, by assuming the “nonprofit starvation cycle” takes the form of donations restricted to use on direct costs and that nonprofit production is Cobb-Douglas in inputs, an increase in indirect costs and savings after a wealth shock imply that nonprofits were constrained by donation restrictions. Proposition 3 shows that if senior employee compensation were a reasonable empirical proxy of private benefits for the nonprofit agent, increases in compensation demonstrate the existence of a principal agent conflict between donors and the nonprofit.

Nonprofits received a cumulative 67% of the average gift size in new donations and government grants (“contributions”) in the two years afterward, giving the nonprofit access to funds totaling 167% of the initial gift. The growth was incremental to the growth in

contributions experienced by the matched control group. As a result, nonprofits spent nearly the entirety of the gift (95%) in the next two years. Compensation comprised 26% (27% of new spending), grant giving comprised 19% (20%), and other costs comprised the remainder.

Importantly, the allocation meant that nonprofits shifted spending from pre-existing operations toward giving the cash away. Recipients increased “on-granting” – giving grants to other organizations or individuals for charitable purposes – by 1.3pp of annual spending concurrent with the gift. Direct service organizations, who on-granted just 0.1% of spending in the year before, increased on-granting by 0.8pp. In the lens of Proposition 1, this shift in nonprofit activities suggests that these organizations have decreasing returns to scale in producing charitable output so that, as organizations grew, the marginal value of dispensing grants outweighed the marginal value of production of core charitable output. While not central to evaluating the consequences of donation restrictions, this finding that nonprofit production is concave is a novel and essential input for normative conclusions about nonprofit market structure and competition.²

I do not find evidence that restricted donations have worsened a “nonprofit starvation cycle” (NSC). The NSC, as modeled, predicts that if nonprofit donations prevented the funding of an input, a wealth shock via an unrestricted donation would loosen the constraint and so spending on the input could increase. As a result, if production were Cobb-Douglas, finding that the relative allocation to indirect costs or savings increased after the gift would show these constraints bound prior to the gift. Despite the gifts’ loosening the restrictiveness of the nonprofit’s donations, recipients did not increase their “overhead ratio” – indirect costs as a fraction of total expenses. The estimates are fairly well-powered. The difference-in-differences estimate implies that average effects that exceed a 0.9pp (5%) increase in the overhead ratio are ruled out at the 95% confidence level since 2019. Expenses relative to liquid reserves and relative to total assets also did not decrease significantly. Average effects exceeding 21% and 18% are ruled out for liquid reserves and for total assets respectively. There was a lack of significant change even for nonprofits whose revenue was more restricted prior to Ms. Scott’s gift and limited evidence that such constraints could be more relevant for smaller recipients.

Finally, I document increases in employment and employee compensation compared to similar nonprofits. Two years after the gift, nonprofits hired 1.4% more employees. Compensation of the chief executive officer (CEO) increases by 5.8%, the wages of the next four highest paid individuals at recipients increased by 11.8%, and employee wages increased by 5.5% two years after the gift. CEO pay increases concentrated in the base pay rather than

²For example, [Bilodeau and Slivinski \(1997\)](#); [Andreoni \(1998\)](#); [Lapointe et al. \(2018\)](#); [Perroni et al. \(2019\)](#) assume increasing returns to scale with a fixed cost paid or donations reached before output is achieved.

bonuses, and a triple-difference between incumbent and newly hired CEO’s suggests only incumbents saw compensation rise. With downward nominal rigidity in wages, this ongoing wage increase was quantitatively large. In present value terms with a 4.5% discount rate, raising compensation corresponded to spending \$0.18 of every dollar of the gift principal and additional contributions on the five highest paid executives, which is similar to the most recent estimate of \$0.17-0.25 for the allocation of tax breaks to executives at for-profit firms (Ohrn, 2023). Because senior management has control over use of funds and their wages increased differentially, I interpret these changes as an upper bound for misallocation from agency frictions in recipients.³

From the funder’s perspective, the results suggest that lifting donation restrictions does not substantially improve the allocation of funds in the way that the “nonprofit starvation cycle” hypothesis would suggest. However, they could be successful in bringing nonprofit wages closer to wages of comparable positions at for-profit firms.

Optimal contract structure for donations: The nonprofit starvation cycle, a principal agent problem, and paternalism. This paper informs a debate amongst funders and academics about the optimal structure for charitable donations and the drivers of this optimal structure. Theories of the nonprofit’s objective function classically highlight consequences of a principal agent problem between the funders and the nonprofit agent who allocates funds. Indeed, Fama and Jensen (1983a) state that “for nonprofits the survival value of such decision systems is due to the assurances they provide that donations are used effectively and not easily appropriated.”⁴ Hansmann (1996)’s canonical justification of nonprofit status is the “nondistribution constraint” that allows nonprofits to commit to their social cause without appropriating funds, echoed by Bilodeau and Slivinski (1998); Glaeser and Shleifer (2001); Ghatak and Mueller (2011); Easley and O’Hara (1983). The threat of appropriation implies that nonprofits hold lower than optimal liquid reserves and under-fund anything that could be misconstrued as perquisites to signal their commitment to service (Jensen and Meckling, 1976; Core et al., 2006; Calabrese, 2011; Fisman and Hubbard, 2003, 2005). This theory is supported by empirical evidence that donations are elastic with respect to the percentage diverted to overhead cost (Gneezy et al., 2014; Meer, 2017; Perroni et al., 2019; Exley, 2020; Hung et al., 2023), which contain many cost items (office, travel, IT) that could be construed as private benefits that do not produce social welfare. Parsa et al. (2022) uses IRS’s 2008 rollout of increased nonprofit governance disclosure requirements to show that donations and overhead ratios become less negatively associated after the onset

³The estimates are an upper bound because higher wages could have other unmeasured benefits to productivity or retention that I do not evaluate in this paper.

⁴Fama and Jensen (1983b, 1985) renew this sentiment and discuss how organization form affects competition between, the survival of, and the investment choices of organizations.

of governance information.

However, an emerging empirical literature suggests that the under-funding of liquid reserves and overhead expenses blunts the nonprofit’s efficient production of charitable activities. Nonprofits do not increase spending during economic downturns as much as donors would hope (Exley et al., 2023), and one key determinant of that phenomenon could be their lack of precautionary savings (Calabrese, 2013). Another could be pressure not to invest in overhead costs, as nonprofits spend less on overhead costs than similar for-profits in the cross-section (Altamimi and Liu, 2022). The welfare impact of increasingly unrestricted giving comes from weighing the threat of misallocation against the under-funding of liquid reserves. Questions about which stakeholder can more easily evaluate the social value created by the nonprofit relate to growing behavioral literature on the impacts and drivers of paternalism (Ambuehl et al., 2021).

I find novel causal empirical evidence inconsistent with this emerging literature. I do not find evidence of the nonprofit starvation cycle because indirect costs and savings do not increase significantly. Instead, I quantify executive compensation increases out of unrestricted gifts that suggest restrictions on donations constrain the principal agent problem. These estimates of executive wage increases are the first quantification in the nonprofit sector out of windfalls, corroborate correlations between donation inflows and executive pay found in Fisman and Hubbard (2005), responsiveness of nonprofit CEO pay to monitoring in Babenko et al. (2026), and causal evidence of windfall spending at for-profit organizations (Blanchard et al., 1994; Bertrand and Mullainathan, 2001; Howell and Brown, 2023; Ohrn, 2023).

Donation crowd in and the nonprofit production function. This paper also validates a corpus of previous work in a new empirical setting that shows that attention drives donations to nonprofits. The theoretical literature has rationalized donation crowd in after a prominent donor’s gift as such gifts acting as an “announcement of support” and hence a positive signal of the charity’s quality (Vesterlund, 2003; Andreoni, 2006; Karlan and List, 2020) or with nonprofits having increasing returns to scale (Andreoni, 1998). The empirical backing for the latter is mixed, where seed money increases giving, but higher match rates do not spur higher giving (Karlan and List, 2007; List, 2011). My model shows that these findings can be reconciled by nonprofits having decreasing returns to scale. A decreasing marginal ability to produce social welfare means that the gift (or matching) not only creates a positive signal but also moves the nonprofit up its production function into lower marginal productivity of additional donations.⁵ Donation crowd in after Scott’s gift

⁵Economies of scale of charitable production is an important input for normative conclusions about the structure of the nonprofit industry (Castaneda et al., 2008; Lapointe et al., 2018), but empirical measurement has been primarily limited to hospitals (see, for example, Gowrisankaran et al. (2015)).

validates prior work on the potency of announcements of support (Kessler, 2017; Hungerman et al., 2026) and demonstrates that even in a setting with very large gifts, the effect of such an announcement overpowers a donation crowd *out* effect from declining returns to scale.

Large gifts. Lastly, this paper is one of the first studies of the spending out of MacKenzie Scott’s gifts and contributes a detailed account and interpretation of how the spending out of unrestricted and unanticipated gifts can shed light on the nonprofit’s objective function. Lee et al. (2023) describes the size of the gifts and spatial distribution of Scott’s giving, some facts that I reproduce here for gifts limited to 2020-2022 excluding universities and hospitals. The Center for Effective Philanthropy performed two surveys of recipients that solicit nonprofits’ plans for how to use their gifts from MacKenzie Scott (Buteau et al., 2021, 2022). Their third report (Arrillaga et al., 2025) also reports revenue, spending, saving, and fundraising by a different sample of recipients alongside the average outcomes for matched untreated nonprofits between 2020-2023. By matching recipients on multiple years of outcomes and separately by cohort then using difference-in-differences, I identify the effect of the gift while mitigating confounding by aggregate time trends. Furthermore, my work quantifies additional outcomes (grant giving, direct and indirect costs, and employee compensation) and reports heterogeneity between nonprofits of different missions. Mayo (2021) uses bequests to study the effect of large gifts on recipients, but my ability to ascertain that gifts were both unrestricted and unanticipated allows me to use them as an exogenous wealth shock to test a model of the nonprofit’s objective function and speak to the nonprofit starvation cycle hypothesis.⁶

The remainder of the paper proceeds as follows. Section 1 presents the model and implications that guide the empirical results. Section 2 describes MacKenzie Scott’s gifts. Section 3 presents the data, variable definitions, and how I construct the sample of unrestricted gifts. Section 4 presents the empirical strategy for identifying the effect of an unrestricted gift of nonprofit recipients. Section 5 presents the results on contribution crowd in and spending, then Section 6 reports how new spending was allocated to test predictions of the model. Section 7 concludes.

⁶The marginal propensity to consume out of large cash windfalls has been studied extensively for consumers, reviewed in Golosov et al. (2024) and to a limited extent at firms. Ohn (2023) reviews literature on tax-related windfalls, (Blanchard et al., 1994; Bertrand and Mullainathan, 2001) use other positive financial shocks, and only von Beschwitz (2018); Howell and Brown (2023) present empirical designs that involve cash shocks exogenous to investment opportunities.

1 Theoretical framework

In this section, I present a model that connects the usage of MacKenzie Scott’s gifts with three questions about nonprofit donations and the nonprofit objective function: 1) Do nonprofits have decreasing returns to scale? 2) Do nonprofits face a starvation cycle that constrains saving and indirect cost spending? And 3) do nonprofits face a principal agent problem in the allocation of funds? I first introduce the model then demonstrate how the setup allows Scott’s gifts to answer each question.⁷

1.1 Setup

Consider a one-period setting where a nonprofit manager can collect donations and spend donations and wealth on behalf of the organization. The manager can allocate spending across four goods: inputs to the production of charitable output $\mathbf{z}_i$, charitable grants m_i , fundraising f_i , and benefits to the manager p_i .

Charitable output is produced by a K dimensional vector of inputs, $\mathbf{z}_i$ via the nonprofit’s own Cobb-Douglas production function $Y(\mathbf{z}_i) = \prod_k z_{ik}^{\phi_k}$ and grants m_i , which are given directly to individuals in need or other organizations. Nonprofits in the same sector have the same production function but can have a nonprofit-specific productivity parameter A_i . The nonprofit produces charity with grants with the expression $b_i m_i$, where b_i is a nonprofit-specific productivity that transforms grants into charitable output. When b_i is small, the nonprofit maximizes charitable output by producing a specific good or service with spending on inputs $\mathbf{z}_i$. When b_i is relatively large, the nonprofit maximizes charitable output by being a grantmaker. The parameterization of charitable output reflects the reality that the value of charity is not *equivalent* to the input spending, but instead produced by it.⁸ Donations D_i are increasing in fundraising spending f_i . The parameter governing the principal’s weight on charitable output over her private utility is $\theta_i \in [0, 1]$.⁹ She has a differentiable, concave utility function $U(p)$. These assumptions amount to the nonprofit agent choosing $\mathbf{z}^*, m^*, p^*$,

⁷In Appendix C.2, I also show this model can provide interpretations for why donations crowd *in* following Scott’s gift. In particular, Scott’s gift is both an announcement of support and an increase in size for an organization with decreasing returns to scale.

⁸For instance, internationally, many estimates of the value of a quality-adjusted life year are larger than the public health intervention required to create it.

⁹The two-pronged motivation of nonprofit manager to maximize charitable output and private benefits follows models of nonprofit behavior in Besley and Ghatak (2005); Babenko et al. (2026) and models of physician behavior in Schnell (2025) and others.

and f^* using Equation (1):

$$\begin{aligned} \mathbf{z}^*, m^*, p^*, f^* = \arg \max_{\mathbf{z}_i, m_i, p_i, f_i \geq 0} & \left\{ (1 - \theta_i) [A_i Y(z_i) + b_i m_i] + \theta_i U(p_i) \right\}, \\ \text{s.t.} \quad W_i + D_i(f_i) - f_i &= \sum_{k=1}^K z_{ik} + m_i + p_i \end{aligned} \quad (1)$$

where $Y(z_i) \equiv \prod_k z_{ik}^{\phi_k}$.¹⁰

1.2 Economies of scale

Do nonprofits have decreasing returns to scale? Proposition 1 shows that the change in the share of spending allocated to grants can directly answer this question for recipients of a gift from MacKenzie Scott.

Proposition 1. *Let $m^*(W)$ denote the nonprofit's optimal choice of grants at wealth W , and define the share of spending "on-granted" as*

$$s_m(W) \equiv \frac{m^*(W)}{z^*(W) + p^*(W) + f^*(W) + m^*(W)}.$$

If on-granting increased in a positive wealth shock $G > 0$,

$$\Delta s_m \equiv s_m(W + G) - s_m(W) > 0$$

and charitable spending, $\sum_{k=1}^K z_{ik}$, was positive prior to the gift, then the production function $Y(\cdot)$ has decreasing returns to scale.

Proposition 1 states that if Scott's recipients had decreasing returns to scale, then the size of their increase in spending would cause them to allocate spending increasingly toward the constant returns to scale charitable output: grant giving. Intuitively, as the nonprofit had declining marginal returns of providing their core good or service, the nonprofit would see higher relative benefits to transferring funds onward to others – directly to organizations or people where the money would go to better use.

The proof (in Appendix C.1) relies on computing Δs_m under various assumptions for the

¹⁰Propositions 1, C.1, 3 do not rely on $Y(\cdot)$ taking a Cobb-Douglas form, but Propositions 2 and C.2 do. This assumption is motivated by empirical results that input shares for both savings and indirect costs remain constant after the wealth shock. If nonprofits face the constraints in Assumption 1 and 4 and production were not Cobb-Douglas, those results suggest a wealth-induced change in input shares was exactly canceled out by any effect of loosened constraints, which is unlikely to hold for multiple outcome variables.

shape of $Y(\cdot)$. The proof reveals that if the relative input share of grants increases after the gift, it must be through an increase on either (1) the intensive margin or (2) the extensive margin. Case (1) can only materialize if a slack non-negativity condition for m remained slack after the gift because Y has decreasing returns to scale. Case (2) can only materialize if the non-negativity constraint bound at W but became slack at $W + G$, a phenomenon which again implies that the production of non-grant goods has decreasing returns to scale. In other words, the nonprofit's production function is quasi-linear in grants, so there are no wealth effects for inputs z^*, p^*, f^* above a minimum wealth threshold. All incremental wealth after the threshold is devoted to the numeraire, m^* , and so its share of spending on m^* out of wealth increases.

1.3 The nonprofit starvation cycle

The nonprofit starvation cycle (NSC) hypothesis asserts that nonprofits are constrained in their ability to save funds for future use and to spend money on indirect costs because donations are earmarked for immediate use on charitable projects. As a result of this constraint, nonprofits must allocate sub-optimally low funds to indirect cost and have lower charitable output. Donors, seeing that lower productivity of their marginal dollar, give less and the nonprofit produces less charity than what they could otherwise without a constraint. The cycle continues because donations are again lower in future periods, and so the nonprofit remains constrained. Restrictions on savings work similarly. The nonprofit's productivity may vary from year to year based on needs of their target population, and even though maximizing the present value of charity means saving some of current donations, the nonprofit cannot do so with restricted donations. Without producing enough charity by having the flexibility to move funds between periods, the nonprofit remains constrained.

In this section, I use the model to design the first empirical tests of this hypothesis using relative spending on indirect cost.¹¹ Specifically, the NSC hypothesis is that donations *constrain the nonprofit's resource allocation* because some donations cannot be used for a given input. The empirical test hinges on finding empirical evidence that this constraint is binding.

Under the framework of my model, restricted donations are equivalent to wealth that is earmarked only for use on input good $z_{j1} \equiv d_i$ termed "direct cost" and not for "indirect costs" $z_{j2} \equiv \iota_i$, fundraising f_i , or private benefits p_i . This restriction in reality takes the form of donations given for specific projects where allocating elsewhere is accompanied by legal or reputational risk. Consequently, the nonprofit starvation cycle hypothesis can be distilled to an additional constraint on the nonprofit objective, Assumption 1.

¹¹Appendix C.2.2 presents the analogous proof for relative allocation of wealth to saving.

Assumption 1. *Fraction $\eta_i > 0$ of donations D_i cannot be used for indirect costs. In other words, the nonprofit also faces a new constraint in its objective function:*

$$\iota_i \leq W_i + (1 - \eta_i)D_i$$

Proposition 2 then shows the empirical content of the nonprofit starvation cycle hypothesis: that with a large unrestricted gift, a constrained nonprofit should weakly increase its indirect cost spending.

Proposition 2. *Define the overhead ratio at a wealth level W as*

$$s_\iota(W) \equiv \frac{\iota^*(W)}{d^*(W) + \iota^*(W)}.$$

When $\theta = 0$, if the change in the overhead ratio is zero,

$$\Delta s_\iota \equiv s_\iota(W + G) - s_\iota(W) = 0,$$

then the “nonprofit starvation cycle” constraint (Assumption 1) was generically slack prior to gift G .^a

^aThe proposition holds subject to a knife edge parameter restriction that is a function of parameters $A, \alpha, \beta, \eta, W, G, b$, and $D(\cdot)$ and defined in the proof.

The “constrained” nonprofit is one where the donation earmark constraint in Assumption 1 binds. The proof makes clear that, under the situation where this constraint binds prior to the gift, spending will increase if the constraint becomes slack after the gift. For the relative spending on indirect costs (overhead ratio) to remain constant, then the NSC constraint must have been slack before and after the gift – in other words, the nonprofit did not have a constraint that is a hallmark of the “nonprofit starvation cycle.”

1.4 Principal agent conflicts

Do nonprofit organizations face principal agent conflicts? Unrestricted donations, by design, cede control from donors to internal stakeholders to direct funds. Proposition 3 shows that, if we can measure an empirical proxy of these benefits p^* , then an increase in that proxy after the gift demonstrates the existence of reallocation.

I use base pay of senior executives as my proxy because (i) directors retain some residual control rights to the firm’s assets and consequently should receive some of benefits p^* and (ii) the assumption that the marginal product of labor of directors is unaffected by the gift

could be more reasonable than the analogous assumption for the CEO.¹² Assumption 2 and 3 formalize these statements.

Assumption 2. *A director’s base pay is the sum of her marginal product of labor (MPL) and potential private benefits.*

$$(\text{Director Salary})_{it} = p_{it} + (\text{Director MPL})_{it}$$

Assumption 3. *Scott’s gift did not change the marginal product of labor of directors:*

$$(\text{Director MPL})_{it}(W_{it}) = (\text{Director MPL})_{it}(W_{it} + G_i), \quad \forall \text{ nonprofits } i. \quad (2)$$

Invoking Assumption 3 for base pay (hereafter referred to as “salary”) is far more reasonable than invoking it for total compensation, which includes incentive pay that could be confounded by other unobserved variables. In particular, this statement rules out “pay for performance” of directors that is reported through salaries rather than bonuses. While Scott’s gift was unexpected, nonprofit managers could have reasonably been rewarded for any fundraising effort that could have been perceived to contribute to their selection as a grantee. Since such rewards are often structured as one-time bonuses, this reward should not appear in base pay, Director Salary_{it}. Assumption 2 is also more compelling for salaries if pay increases were due to financial constraints. Howell and Brown (2023) posits that financially distressed firms borrow short-term from employees then pay back after a cash windfall. Assumption 2 precludes the possibility of repayment through salary increases. As salaries are downwardly rigid, this repayment would more reasonably occur through bonuses.

Proposition 3 then directly follows from the two assumptions and shows principal agent conflicts can be detected in the wake of Scott’s gift.¹³

Proposition 3. *Under Assumptions 2 and 3, if director base pay increases after a wealth shock, then $\theta_i > 0$.*

The proof is straightforward and detailed below.

Proof of Proposition 3. Because of Assumption 2, the change in a director’s salary after Ms.

¹²For example, Gabaix and Landier (2008) present a model where CEO pay scales with organization size, which did grow due to Scott’s gifts. However, later empirical comparisons between relative compensation increases for new and existing CEO’s suggest that this mechanism is not a first order determinant of nonprofit CEO compensation.

¹³The result can also be generalized to a circumstance where the directors’ marginal products of labor can be successfully controlled for using observable variables, such as the size or sector of the organization.

Scott’s gift can be written as:

$$\begin{aligned} \text{Director Salary}_{it}^* (W_{it} + G_i) - \text{Director Salary}_{it}^* (W_{jt}) &= [p_{it}^* (W_{it} + G_i) - p_{it}^* (W_{it})] \\ &+ [(\text{Director MPL})_{it} (W_{it} + G_i) - (\text{Director MPL})_{it} (W_{it})] \end{aligned} \quad (3)$$

Assume $\theta_i = 0$. Then $p^*(W) = p^*(W') = 0$, so the first term is zero. By Assumption 3, the second term is also zero, so the change in director salary with a gift G is zero. This is the contrapositive of what we’d like to prove because the support of θ_i is $[0, 1]$. So, if director wages do change after a wealth shock, then $\theta_i \in (0, 1]$, which means $\theta_i > 0$. ■

2 Setting

The size and unanticipated nature of MacKenzie Scott’s gifts to hundreds of nonprofits between 2020 and 2022 make these gifts an ideal setting in which to study the impact of unrestricted grants. News reports and Ms. Scott’s blog posts corroborate that nonprofits were unaware of the gift until shortly before receiving it. Grants were targeted to large organizations in a charitable sector where spending is highly concentrated. I estimate 88% of spending in 2019 was done by organizations of at least the size of Scott’s median recipient.

2.1 “Transformative” gifts

Since 2019, philanthropist MacKenzie Scott has given \$25B to over 2,700 nonprofits (Chung, 2026). These gifts have been large on an absolute and relative scale. Figure 1 shows the distribution of relative gift sizes for my sample of 2020-2022 gifts. This shows that the average gift received was \$8.4M and 87% of all revenue the recipient had received in 2019. Scott’s gifts are often referred to as a “game-changer”, “significant”, and “transformative” for recipients (Meals on Wheels America, 2020; Turtinen, 2022; Nash, 2023; Hired, 2024). Appendix Figure A.2 shows that these gifts focused on “Direct Service Organizations” – food, housing, and human services organizations like the YMCA, Communities in Schools, and Meals on Wheels that spend money to provide goods and services directly to individuals in need. Other recipients were philanthropy support or “grantmaker” organizations who primarily collect donations then distribute them via grants within particular specialty areas, such as LiftFund, the Hispanic Scholarship Fund, or the Pittsburgh Foundation. For 88% of nonprofits in a separate sample of survey respondents, her gift was the largest unrestricted gift they had ever received (Buteau et al., 2021).

2.2 MacKenzie Scott’s gifts were unrestricted and unanticipated

The chosen sample of MacKenzie Scott’s gifts allow me to assess the impact of a large, unrestricted gift on a nonprofit recipient because her gifts were unanticipated and not fundraised for. Ms. Scott has been public about the lack of restrictions associated with her gifts. In announcing one round of gifts, she mirrors the language of the “nonprofit starvation cycle”. She states, “not only are non-profits chronically underfunded, they are also chronically diverted from their work by fundraising, and by burdensome reporting requirements that donors often place on them.” As a result, “the entire commitment would be paid upfront and left unrestricted in order to provide them with maximum flexibility” (Scott, 2020b). Specifically, the team “welcomed them to spend the funding on whatever they believe best serves their efforts.”

Her process between 2019-2022 also left gifts unanticipated. She describes a process of “quiet research” culminating in sharing the gift with the nonprofit leadership “for the first time over the phone”, specifically with the goal to “give them an immediate gift for use however they choose.” Buteau et al. (2021) reports that only 44% had an interview with Scott’s team, and 28% had to provide financial documents. Despite the possibility that some of the recipients had some advance notice, news reports emphasize the shock of nonprofit executives because of the short vetting process. Reporting on these gifts include quotations such as:

- *“When I first received an email, it just said, “There’s a donor who’s potentially interested in giving money to the college. Would you have time to have a conversation with me?” And that’s really all it said... So it was kind of, Is this real? Is this fake? What’s the deal? But I did have a conversation with somebody doing some research for [Scott]. By the end of the conversation, she basically said that MacKenzie would be giving us \$8 million.”* (Olmstead, 2021)
- *“In a seemingly random act of kindness, a nonprofit C.E.O. receives an email from one of Scott’s aides”* (Schleifer, 2023)
- *“The gifts routinely shock recipients.”* Chung (2026)

That shock is important to identification because it means that executives had little time and inclination to spend out of the gift before receiving it. Panel data can identify the effect on nonprofits of the gift as long as the date at which they began to act on the gift was simultaneous with the date that MacKenzie Scott reported the gift as being given. Appendix Figure A.3 shows that the time series of key outcome variables for recipients and their matched control nonprofits look similar in the years prior to the gift then experience

a noticeable break in trend. Lastly, unlike most previous documentation of large gifts, the gifts were not fundraised for. The benefit of this lack of fundraising is that spending should not be subject to a flypaper effect. Consequently, resources have no *a priori* reason to flow to fundraisers, and the gift is a windfall whose funds could be directed anywhere in the organization.

The primary limitation of this study’s ability to measure the impact of large, unrestricted gifts on 501(c)(3)’s is the endogeneity of the choice of recipients. While precise selection criteria have been kept opaque to prevent fundraising for her gifts, Ms. Scott describes the diligence process as “data-driven and rigorous.” Selected nonprofits have “high potential for sustained positive impact, including stable finances, multi-year track records, measurement and evidence of outcomes, and experienced leadership representative of the community served” (Scott, 2020b). This assignment mechanism had at least two measurable implications. First, the recipients are far larger organizations than the median 501(c)(3) (Table 1). Their size could be a result of higher productivity and could make their responses to a cash windfall non-representative of the full population of charitable organizations. Second, recipient nonprofits skew toward human services and grantmaking (Appendix Figure A.2). Consequently, I match each recipient to an untreated nonprofit and assume the treated nonprofits would have had parallel trends in their outcome variables to the untreated nonprofits to identify their counterfactual path of spending and saving, after controlling for sector-by-time fixed effects. Details on this approach are left to Section 4.

Importantly, while the empirical design speaks only to the average effect on treated organizations, the size of the recipients strengthens the case that the gifts were representative of how the median dollar would be spent out of a windfall in the charitable sector. The median annual spending of recipients was \$7.9M in 2019, the last year prior to treatment. In 2019, 88% of charitable spending was done by organizations of at least that size.¹⁴

The sample stops at 2022 because March 2023 marked a change to Scott’s giving strategy. In March 2023, Scott announced that \$250M would be awarded through an Open Call that allowed nonprofits to apply for funding. The change brings concerns that nonprofits who received Open Call or subsequent funding could act differently than those who received an unanticipated gift, and hence the analysis will focus on a subsample of early recipients.

¹⁴This statistic uses the spending of Form 990 filers whose expenses exceed the median recipient’s expenses, \$7.9M, divided by Hodge (2024)’s 2019 estimate of total 501(c)(3) spending. Using only the spending of Form 990 filers, the statistic is 91%.

3 Data

This paper uses MacKenzie Scott’s public disclosures on her unrestricted giving on YieldGiving and financial information of U.S. nonprofits from the IRS Form 990 tax returns posted online by the Internal Revenue Service in order to identify the gift recipients and then the impact of the unrestricted grant on their finances.

3.1 Observing the behavior of nonprofits

The Form 990 provides a detailed annual financial record of each nonprofit’s activities that will reveal the impact of unrestricted gifts on nonprofit finances. I use two sources of Form 990 filings.

I use 2012-2023 annual Form 990 microdata extracts on IRS.gov for most financial variables. All firms tax-exempt under Section 501(c)(3) with gross receipts exceeding \$200,000 or assets exceeding \$500,000 during their fiscal year are required to file a Form 990 to the Internal Revenue Service. These extracts report each nonprofit’s revenue, expenses, assets, and liabilities by fiscal year. Revenue is broken down by source, including charitable contributions and program service revenue – revenue generated from providing goods and services for their tax-exempt activity.¹⁵ Contributions are the sum of government grants and donations that come from all other sources, such as grants from foundations, individuals, and corporations. Expenses are broken down by source, including grants provided to individuals, grants provided to organizations and governments domestically and abroad, employee salaries, employee benefits, pension contributions, office expenses, IT expenses, and travel expenses. The breakdown of assets and liabilities has similar detail, including non-interest bearing savings, public and private securities investments, land, building, and equipment capital stock, depreciation, and secured and unsecured loans. Nonprofits also report the number of individuals employed in that year per their W-3.¹⁶ I define employee compensation as the total compensation for workers at the nonprofit. It is the sum of several reported items on the Form 990: the compensation of “directors, officers, and key employees”, “other salaries and wages”, “other employee benefits”, pension plan accruals and contributions, and payroll taxes. The average compensation per non-senior employee is the quotient of the total salaries plus employee benefits and pension contributions divided by employees minus the number of individuals with positive wages on this list.

I supplement this panel with the full Form 990 for electronic filers (“EFilers”), whose full

¹⁵For instance, the Goodwill has significant program service revenue from selling clothing.

¹⁶I do not use these estimates for any nonprofit classified as with NTEE code Q, “International, Foreign Affairs, & National Security”, because many of these organizations have multiple locations outside the United States whose employees need not be reported on W-3 forms.

tax returns are posted on IRS.gov in full for fiscal years ending from 2016 to 2025. These are digitally available on IRS.gov only when a nonprofit has filed their tax-return electronically in the year. Filings are added periodically throughout the year as they are received, and the panel includes filings uploaded on or before March 10, 2026. Electronic filing was voluntary but common for Form 990 filers, then became required for all tax years ending after July 31, 2020. Efiler returns provide additional compensation information, employee counts, and additional detail on the sources of nonprofit revenues, and the allocation of expenses into Program Service, Management and General, and Fundraising. Program Service spending is spending directly related to the tax-exempt “program” of the nonprofit. Following the prior literature including [Altamimi and Liu \(2022\)](#), the overhead ratio is defined as one minus the fraction of spending assigned to “Program Service” in the year.

These full returns provide the compensation of each of their “directors, officers, and key employees.”¹⁷ Using this list, I report the compensation of the highest paid employee, whom I assume is the CEO. I compute the average director compensation as the average total compensation of the second through fifth highest paid employees on this list. In testing the robustness of compensation increases, I use base and bonus compensation as defined in the Schedule J for nonprofits that provide this detail. When testing predictions of the nonprofit starvation cycle, I compute the expense to asset ratio as the total annual expenses divided by the total assets at the end of the year prior. I compute the expense to liquid asset ratio by limiting assets to the value of non-interest bearing savings, interest bearing savings, and publicly traded securities only.

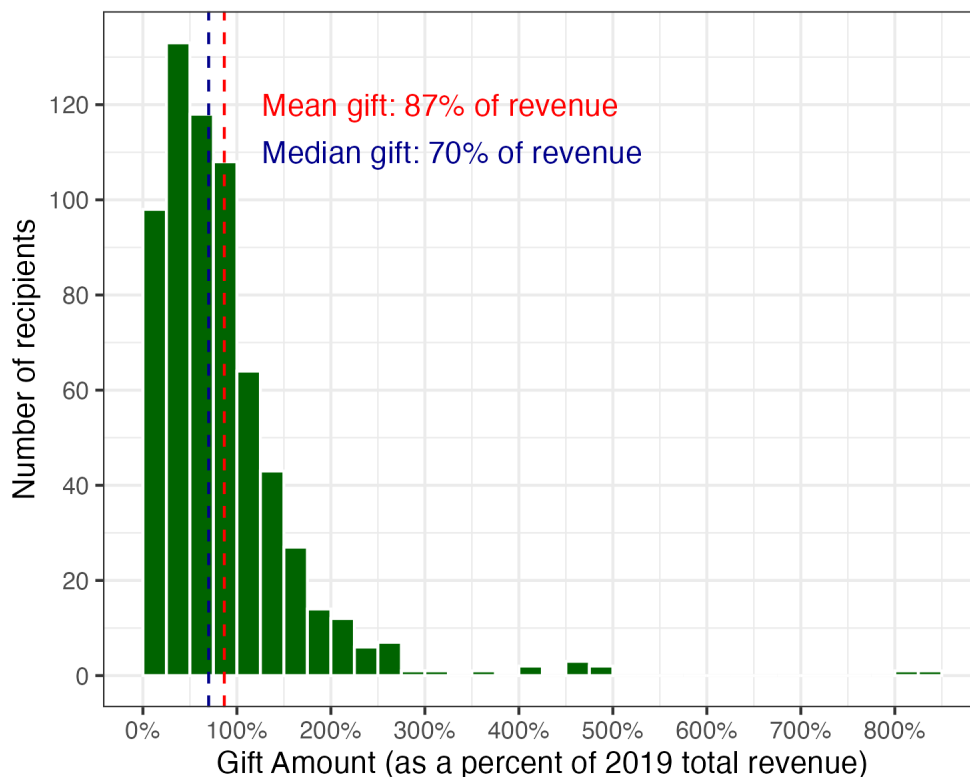
In some cases, I show results separately for grant making organizations and direct service organizations (DSO’s). This classification, common in the literature, distinguishes nonprofits who give cash grants to other organizations from nonprofits who provide goods or services directly. Quantitatively, I define DSO’s as nonprofits whose grant funding comprises 0.7% or less of their annual spending, where this is the median share of spending given as grants in the year prior to the gift. The remaining recipients are grantmakers.

In order to have industry codes for all treated and control nonprofits, I limit my population of 501(c)(3) organizations to organizations that survived through November 2023 using the Exempt Organizations Business Master File (EO BMF). The National Center for Charitable Statistics (NCCS) uses the National Taxonomy of Exempt Entities (NTEE) classification system to divide nonprofits into 26 nonprofit groups, similar to a NAICS code for for-profit firms, and the IRS provides this classification publicly only through the routinely updated EO BMF. The resulting dataset of nonprofits is an unbalanced panel of 234K

¹⁷Inclusion in this list of executives is discretionary so the list may not be entirely exhaustive. The salaries of omitted executives would be part of the sum of non-senior employees salaries.

nonprofits and constitutes the Form 990 population to which I merge in unrestricted gift information.

Figure 1: *Distribution of gift amounts relative to recipient size*



Note: This figure shows the distribution of MacKenzie Scott’s gifts as a fraction of the total revenue of recipients in 2019.

3.2 Identifying unrestricted gifts

My sample of unrestricted gifts is gifts listed on MacKenzie Scott’s Yield Giving’s website that were directed to the entire organization and disclosed a gift amount between 2019-2022, with a handful of other limitations. Yield Giving’s website provides the only comprehensive list of recipient organizations names paired with the year and size of each unrestricted gift and forms my sample. This list of donations provides the name, service area, and URL of the organization alongside the calendar year of donation. 27% of gifts posted on Yield Giving as of January 2023 did not list a gift amount and were consequently excluded from the sample.

I further restricted the recipient population to nonprofits who could be matched to my Form 990 population. Using the SOI Tax Stats tool on IRS.gov, I looked up the Employer

Identification Number (EIN) for each of MacKenzie Scott’s recipients on Yield Giving that allows me to find their Form 990. When these matches were ambiguous, I confirmed the match using URLs listed on Yield Giving with Guidestar Pro, which provides EINs, names, and URLs for this population of 501(c)(3) organizations. I dropped all gifts given to particular funds within organizations because these gifts were unlikely to be unrestricted for use across the organization.

Using these NTEE codes, I exclude gifts that MacKenzie Scott gave to churches, hospitals, and universities, since these organizations frequently have multiple subsidiaries that make tracking the full finances of these organizations challenging. I limit the population of untreated 501(c)(3) organizations used for this analysis to those that were active and classified by NTEE as of November 2023 using the NCCS’s Nonprofit Masterfile so that I limit analysis to nonprofits that survived through the intervention to be comparable to the treated nonprofits.¹⁸ In order to report effects on employee compensation, I limit the sample to recipients with more than two employees. This focus limits the sample to 643 nonprofits who received funding from MacKenzie Scott between 2020 and 2022.¹⁹ The treated nonprofits were far larger in assets and expenses, more reliant on contributions, and had higher average wages for employees, directors, and CEOs (Table 1, panels 1 and 3).

I validate the timing of the gift using MacKenzie Scott’s blog dates, news articles, and gifts of the same size from tax-exempt organizations linked to her. Many nonprofits have fiscal years that end after their calendar year, and this ambiguity prevents the direct match of Yield Giving calendar year dates and the fiscal year that the gifts were realized in Form 990’s. I ameliorate this ambiguity in two ways. MacKenzie Scott announced three waves of gifts via blogposts (Scott, 2020a,b, 2022b,a). I use the precise dates of the blog posts, when available, to ensure that the gift falls in the correct fiscal year. Next, I use the full universe of grants reported in all Form 990 Schedule I’s and Private Foundation returns (Form 990-PF) between 2019-2023 to find gifts that matched Scott’s in size. News coverage has cited Scott’s transfer of funds through other tax-exempt organizations, such as a DAF through Fidelity Charitable, the Chicago Community Trust, and the National Philanthropic Trust (Schleifer, 2023).²⁰ Consistent with this reporting, I match the majority of my 643 gifts to a gift from one of these organizations, as well as the Silicon Valley Community Foundation, and use their fiscal date to update treatment timing. As shown in Appendix Figure A.1, the final

¹⁸Interesting further analysis of these gifts could examine how the gifts changed the survival of other nonprofits, but this is not the focus of this paper.

¹⁹While Yield Giving lists six gifts given in 2019, none survive the sample restrictions because four do not list a gift amount and two are given to specific projects within the organization.

²⁰At this step, I drop the National Philanthropic Trust and the Bridgespan Group from the sample of gifts, as they were publicly reported as partners in selecting gifts and so the timing of their receipt of funds is unknown.

sample of gifts range in size from less than \$1M to \$50M.

3.3 Calculating exposure to restricted donations

How nonprofits use unrestricted windfall donations could depend on how constrained they were prior to the windfall. As a result, I study the dosage effect of restrictions on their revenues by defining a measure of a nonprofit’s “restrictedness”. I define the nonprofit i ’s “restrictedness” in a given year t as the dollars of government grants and restricted foundation grants D_{fit} from foundations f that the nonprofit receives divided by its total revenue.

$$R_{it} = \frac{\text{Government Grants}_{it} + \sum_f D_{fit}^{\text{Restricted}}}{\text{Total Revenue}_{it}} \quad (4)$$

A higher percent of revenue that comes from restricted revenue sources corresponds to more constraints on the nonprofit’s ability to fund indirect expenses and save. The Efilers dataset lists contributions coming from government grants for each nonprofit. Because I cannot see government-grant-level information in Form 990 data, I assumed that government grants are all restricted funds based on the fact that the typical award requires an application to meet a specific request and standardized follow-on reporting ([U.S. Office of Management and Budget, 2024](#)). Schedule I lists every grant a nonprofit i received from a filing foundation annually with a stated purpose reported by the foundation. I classified grants as unrestricted if the stated purpose included the terms “Unrestricted” or the more formal terms used in the industry “General Support,” “General Operating Support,” “General Use,” “General Purpose,” “General Fund,” “General Contribution,” and their derivatives. The remaining grants were assumed to be restricted gifts, $D_{fit}^{\text{Restricted}}$.

4 Empirical Strategy

To identify the impact of the gift on each recipient, I match each recipient to four financially comparable nonprofits. The recipient sample and comparable nonprofits move in parallel between 2012 and receipt of the gift. With the assumption of parallel trends in the post-period supported by this pre-period co-movement, I can use difference-in-differences to identify the average effect of the gift on the 643 recipient nonprofits.

4.1 Matching recipients to a comparable nonprofit

Because gifts were not necessarily assigned randomly to recipients, I use a matched sample to estimate all my results. I match each recipient of a gift from MacKenzie Scott to four

additional 501(c)(3)’s using nearest-neighbors-matching. Using expenses, grants given, donations received annually, and number of employees in the three years prior to the gift, I find four non-recipient nonprofits for every recipient nonprofit. The intent of matching on multiple pre-period years stems from the fact that recipients were faster growing than the average 501(c)(3), and so matched controls have similarly larger growth rates than if matching on a single year of financial data. The resulting matched 501(c)(3) organizations are closer in activity and closer in key financial variables to the treated nonprofits than to the full population of 501(c)(3) organizations (Table 1, panels 1 and 2). Appendix Figure A.3 shows that the mean outcomes of the matched sample also mirror the pre-period trend in key variables in the years prior to MacKenzie Scott’s gifts.

Table 1: *This table lists the unweighted summary statistics of my sample of nonprofits who received a gift from MacKenzie Scott compared to the full set of 501(c)(3) charities and the matched control nonprofits in 2019. Panel 1 shows statistics for the sample of MacKenzie Scott gift recipients in 2019. Panel 3 shows that the population of Form 990 filers in 2019 were far smaller on key outcome variables than the sample of Scott’s recipients. Panel 2 shows summary statistics for the matched control sample in 2019. The control nonprofits were selected with nearest neighbors matching on expenses, donations, grants given, and number of employees in the three years prior to receiving a gift. Nonprofits targeted by MacKenzie Scott were far larger in employee count, but the matched control is closer than the full population in other outcomes of interest.*

	Treated		Matched control				All Form 990 501(c)(3)’s			
	Mean	Std. Dev.	Mean	Std. Dev.	Diff in Means	Std. Err.	Mean	Std. Dev.	Diff in Means	Std. Err.
Assets (\$M)	37.1	106.9	53.3	260.3	-16.2*	6.6	8.1	130.0	29.0***	4.2
Donations and Grants (\$M)	18.5	112.5	16.1	60.6	2.4	4.6	1.9	35.6	16.6***	4.4
Expenses (\$M)	24.9	116.5	22.5	77.5	2.4	4.8	3.5	44.0	21.4***	4.6
Employees	367.7	912.3	238.5	604.0	129.2***	37.9	47.8	282.4	319.9***	36.0
Avg. Employee Comp. (\$K)	43.5	31.7	41.9	30.2	1.5	1.4	29.8	31.8	13.7***	1.3
Avg. Director Comp. (\$K)	97.7	102.1	103.7	139.5	-5.9	5.1	26.0	98.4	71.7***	4.2
Highest Comp. (\$K)	240.5	193.0	258.7	321.3	-18.2+	10.1	99.5	311.4	141.1***	7.8

4.2 Identifying the effect of gifts on recipients

Matching the recipient sample to financially comparable, untreated nonprofits combined with assumptions about the data generating process allow for the interpretation of regression estimates on the resulting sample as the average causal effect of MacKenzie Scott’s gift on treated nonprofits. For any given outcome Y_{it}^c for a nonprofit i at year t assigned to cohort year of treatment c , the parameter of interest is the average effect of the gift k years after it was received, or the *average treatment effect on the treated*. Denote this parameter by

$ATT_Y(k)$. In my baseline results, I make three assumptions in order to identify $ATT_Y(k)$: (1) that the treated and untreated nonprofits would have had parallel trends absent treatment, (2) that the gifts were unanticipated, and (3) that treatment effects were homogeneous between cohorts c . The formal statement of each assumption can be found in (Sun and Abraham, 2021).

Armed with these assumptions θ_k from the following regression estimates $ATT_Y(k)$

$$Y_{it}^c = \alpha_i + \delta_t + \sum_{k=k_{\min}, k \neq -1}^{k_{\max}} \theta_k 1(t - c = k, D_i = 1) + \varepsilon_{it}. \quad (5)$$

The assumptions rule out the possibility that pre-treatment periods $k < 0$ could contaminate the estimate of the dynamic treatment effect θ_k . Matching recipients to an untreated control group that is observably similar prior to treatment is intended to bolster the credibility of Assumption (1). That said, Assumption (1) could still be violated by residual time varying differences in trend between the treated and matched control in the post-period. One salient example of such a violation would be if COVID-19 related shutdowns affected the types of nonprofits that were treated differently from the types of nonprofits that were untreated. To mitigate this concern, all empirical specifications include nonprofit sector (1-digit NTEE code) by year fixed effects, so the residual variation would have to affect nonprofits differently within sector. If nonprofits anticipated the gift, then matching would distort the selection of a control that meets the parallel trends assumption. News reports that the gifts were unanticipated by recipients (Section 2) corroborate Assumption (2). To lend credence to Assumption (3) in the empirical setting, I use the interacted event study regression offered by Sun and Abraham (2021) to estimate $ATT(k)$ to test my baseline result on donation crowd in. That specification is robust to relaxing Assumption (3) and produces very similar results.

In empirical results that measure an outcome Y_{it} in dollars, I weight regressions by inverse average pre-period revenue of nonprofits to correct for heteroskedasticity to improve the precision of my estimates (Solon et al., 2015). As implied by Table 1, recipients varied greatly in size, as measured in revenue. Since nonprofit total spending closely matches annual revenue and many outcomes Y_{it} are portions of total spending, heteroskedasticity that stems from differences in revenue would skew results toward reflecting the changes in spending of the largest recipients rather than the average.

In order to measure the percent of the average gift that outcome Y_{it} comprises, I normalize treatment by the average size of the gift in each fiscal year cohort c , G^c , similar to recent

work estimating consumers' marginal propensity to consume (such as [Goloso et al. \(2024\)](#)):

$$Y_{it}^c = \alpha_i + \delta_t + \sum_{k=k_{\min}, k \neq -1}^{k_{\max}} \tau_k G^c \mathbf{1}(t - c = k, D_i = 1) + \varepsilon_{it}. \quad (6)$$

This can be read as the share consumed on Y_{it} out of the average gift. However, since observations are weighted by inverse size in the regression but G^c is an unweighted average, running Equation (6) would bias down an estimate of τ_k . I instead compute the average gift within size quartile then run a fully interacted version of Equation (6) to collect a τ_k for each recipient size quartile. Using the delta method, I then compute a final average τ_k . Appendix B explains this approach in more detail.

4.3 Identifying the differential effect of restricted revenue on recipients

In order to evaluate predictions about restricted giving, Section 6.2 tests for larger treatment effects based on how restricted the nonprofit was prior to the receipt of MacKenzie Scott's unrestricted gift. The specification used for these tests is:

$$Y_{it}^c = \alpha_i + \delta_t + \beta \mathbf{1}(t \geq c, D_i = 1) + \phi \bar{R}_i \mathbf{1}(t \geq c, D_i = 1) + \varepsilon_{it} \quad (7)$$

where $\bar{R}_i$ is the average restrictedness of recipients prior to receiving a gift from MacKenzie Scott, calculated as in Section 3.3. A significant ϕ is interpreted as the additional treatment effect from Scott's gift on restricted nonprofits for nonprofits with 100% more restrictedness.

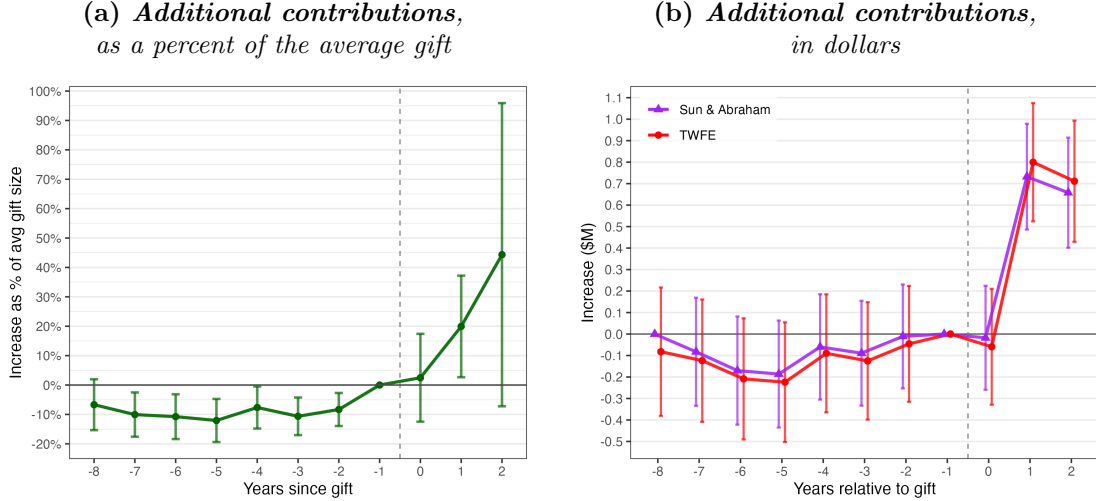
5 Adding up the impact of the cash windfall: contribution crowd in and spending

I find that the size of the cash windfall is 167% of the original gift amount because nonprofits receive 67% of the average gift amount in additional contributions: donations from foundations, individuals, and corporations and grants from governments. Funded by these additional contributions, nonprofits save the gift then increase their annual expenditures so that they spend the equivalent of the entirety of the gift principal within two years.

5.1 Additional contributions

Nonprofits received 67% of the average gift in additional contributions and grants in the first two years after the gift. I find heterogeneity between grantmakers and direct service

Figure 2: Additional contributions to recipient nonprofits



Note: (a) This figure shows the estimate of τ_k computed by the delta method from the interacted regression construction of Equation (6) for total contributions (donations and grants) reported incremental to MacKenzie Scott's gift, measured as a percent of the average gift (\$8.4M). Ms. Scott's gift coincided with an increase in donations and grants from other individuals and organizations relative to untreated nonprofits. (b) This figure shows the estimate of τ_k from Equation (5) for contributions to recipient nonprofits using the two-way-fixed effects difference-in-differences specification and the weighting specification suggested by Sun and Abraham (2021). Point estimates in dollars of new contributions for treated organizations are quantitatively similar. The X-axis tracks years k since the gift was received. Standard errors are clustered at the matching group level. Error bars show 95% confidence intervals.

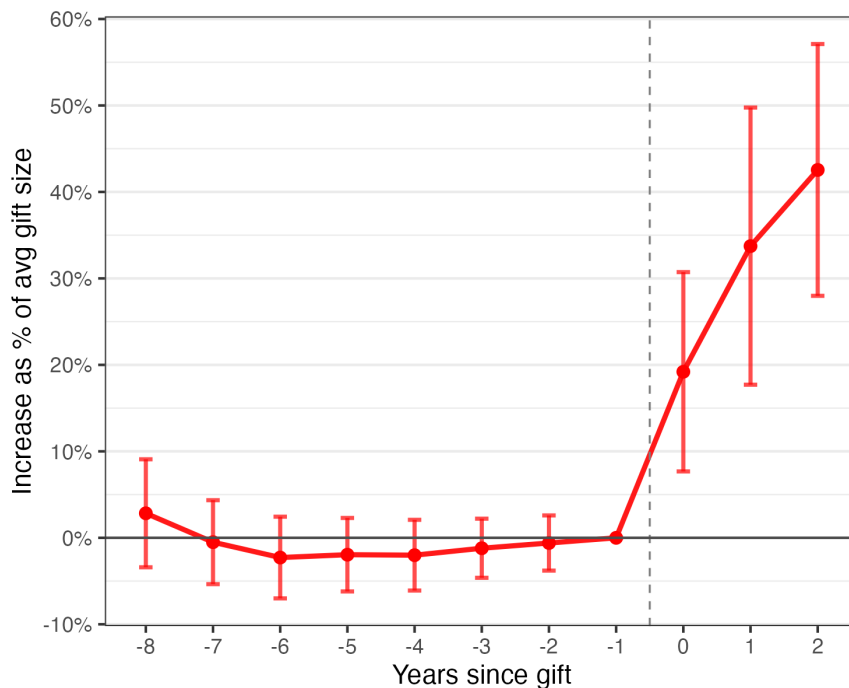
organizations (DSO)'s in the types of contributions that crowd in to the recipient. Concurrent increases in fundraising expenses explain only a small portion of the increase.

Figure 2a reports τ_k from Equation (6) for $k = -8$ to $k = 2$ years after receiving a gift from MacKenzie Scott. These results suggest that, in the year of MacKenzie Scott's gift, treated nonprofits received an average of 2% of the average gift (\$8.4M) concurrently from the combination of (1) donations from other individuals and organizations and (2) government grants. This rose to 20% and 44% of the gift for the next two years, tallying to 67% of the original gift.

Treatment effect heterogeneity could also lead to contamination bias in the measurement of τ_k in Figure 2a. However, reproducing Figure 2a with the estimator suggested by Sun and Abraham (2021) would require creating weights with both quartiles and cohorts or repeating the regression for each quartile. Due to the relatively small sample size and the need to estimate twelve sets of 1-digit NTEE by time fixed effects, I instead compare Sun and Abraham (2021)'s specification with Equation (5). Figure 2b shows that the results are quantitatively similar. Furthermore, crowd in of donations, measured in dollars, is statistically significant

at the 95% confidence level one and two years after receiving a gift unlike in the specification in Figure 2a. Appendix Figure A.4 shows that these contributions derived from both government grants and new donations from other sources. Appendix Figure A.5 shows the effect on grantmaking organizations and DSO’s separately.

Figure 3: *Increase in dollars spent*



Note: This figure shows total spending out of the gift, as a fraction of average gift size. The X-axis tracks years k since the gift was received. The Y-axis shows the estimate of τ_k from Equation (6) for expenses relative to gift year, with 95% confidence intervals. The point estimates imply that the gift coincided with an increase in expenses equivalent to \$0.19 per dollar of the average gift, rising to \$0.34, \$0.43 after the gift. In other words, two years after the gift, recipients had spent the entirety of their gift, on average.

Weakly correlated changes in fundraising suggest that the gift was primarily a donation supply shock. I compute the correlation between the percent increase in contribution and percent increase in fundraising at treated organizations, from their average levels before and after the gift. The correlation between the change in fundraising and contributions is only 0.17 – meaningful, but relatively modest. If fundraising is the input good to “producing” donations, then the marginal donations collected per dollar of fundraising should have been roughly unchanged and fundraising and donations should be highly correlated. Instead, it is much less correlated than spending, and I conclude that while fundraising contributed to the contribution crowd in, it was likely not the primary driver.

5.2 Total spending

Concurrent with additional donations, nonprofits increased spending in the two years after the gift and spent the entirety of the gift in the first two years. Figure 3 plots the fraction of the average gift by which nonprofits increased annual expenditures. I find that annual spending increased steadily starting in the year of the gift from 19% in the year of receipt to 34% and 43% one and two years after the gift. The cumulative effect of this spending is equivalent to spending 95% of the gift within two years.

To investigate whether the growth in spending was associated with the donation crowd in, I compute the correlation between the percent increase in donations and in spending by recipients as done for fundraising. The correlation between the increase in spending and contributions is 0.34. The stronger correlation in growth (unlike the relationship between fundraising and donations) suggests that nonprofits chose how much they expanded based on the crowd in they received. The one-time gift was a starting point, but contribution crowd in played a role in the resulting size of the nonprofit.

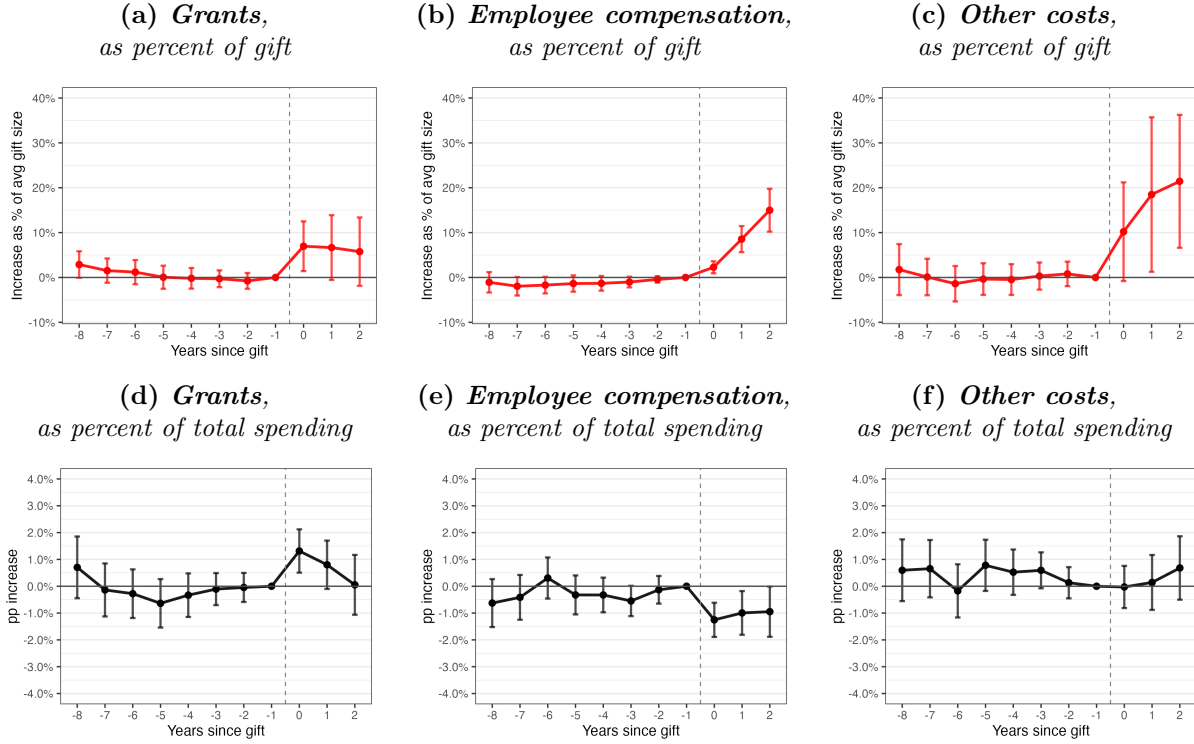
6 Allocation of spending

By two years after the gift, 19% of the average gift was spent on grants to individuals and other organizations, 26% was spent on compensation, and 50% was spent on other costs. In relative terms, while grant giving comprised more of spending than prior to the gift, other spending remained closely in line with pre-existing allocations. I discuss how the results square with empirical tests suggested by the model in Section 1.

6.1 Grant giving

Figure 4 shows the allocation of spending to grant giving by nonprofits in two ways. In Figure 4a, grant giving comprises \$0.07, \$0.07, and \$0.06 per dollar of the average gift, reporting τ_k from Equation (6). Figure 4d shows that this increase was more than proportional to the spending that recipients had on grant giving prior to Ms. Scott’s gift. It reports the binary treatment effect of the gift, θ_k from Equation (5), on the percent of annual spending given as grants to other organizations – 501(c)(3)’s or others – and individuals. Examples from this set of recipients include a food bank transferring funds to churches, school districts, and other food banks. In the year of the gift, nonprofits allocated 1.3pp more of spending toward giving grants. This measure is the total dollar amount given in grants and does not include any supporting costs of selecting grantees and administering the grant. In other words, this spending is the dollars of cash given directly in the tax-exempt purpose either to

Figure 4: Allocation of spending



Note: (a, b, c) The panels show the estimates of τ_k from Equation (6). As treated nonprofits increased their annual spending, they spent more money on grants, compensation, and all other costs. (d, e, f) Grants became 1.3pp more of total spending in the year of the gift. Compensation reduced commensurately, and other expenses remained proportional to their previous amounts. The Y-axis range is the smallest interquartile range of these three fractions (percent of expenses represented by grants).

related organizations or to the population to whom the nonprofit typically provides goods and services.

Appendix Figure A.7 shows this change in allocation is observable especially in DSO's, who give away on average less than 0.1pp of their spending in the year prior to the gift. Concurrent with the gift, 0.8pp more of DSO spending went to grants (Figure A.7b), which more than doubles the percent of annual spending devoted to grants. In contrast, the change in the relative allocation toward grants was smaller for grantmakers, who were already spending regularly on grants. Applying Proposition 1, the change in allocation implies that DSO recipients on average had decreasing returns to scale in charitable production. Nonprofits were eager to spend the money, so spending steered toward objectives, like grantmaking, that had higher near-term marginal returns. The empirical specification measures the average effect on treated nonprofits and cannot speak to how a windfall would affect the average nonprofit. In speculating how results extrapolate to this case, the selection of nonprofits to have "high

Table 2: This table shows the outcome of Equation (7) for overhead ratio, expense to liquid asset ratio, and expense to asset ratio (columns (I), (III), (V)). Applying Propositions 2 and C.2, if recipients had suboptimal overhead ratios or too large expense to asset ratios due to donation restrictions, then a large unrestricted gift should change their level. No coefficients in those specifications are significant. Furthermore, the gift should affect nonprofits with higher restricted donations more, and so columns (II), (IV), and (VI) interact receiving a gift with the fraction of prior revenues that were restricted. This coefficient on this new term is also not significant.

	Overhead ratio		Exp.to liquid asset ratio		Exp. to total asset ratio	
	(I)	(II)	(III)	(IV)	(V)	(VI)
Post	0.00387 (0.00261)	0.00175 (0.00373)	-0.785 (2.167)	-1.15 (2.81)	-0.0535 (0.1245)	-0.137 (0.165)
Post $\times$ (Percent of rev. restricted)		0.00771 (0.01095)		1.37 (5.25)		0.313 (0.415)
Num.Obs.	21230	21223	21633	21626	33660	33649
R2	0.820	0.820	0.195	0.195	0.466	0.466

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

potential for sustained positive impact” suggest that Scott’s recipients had an even greater incentive to keep funds for productive future use within the firm, and so the average effect of a windfall could, in reality, be even larger.

6.2 Testing predictions about unrestricted giving: Indirect costs and saving

In this section, I use Propositions 2 and C.2 to test for the presence of the nonprofit starvation cycle in recipients of gifts from MacKenzie Scott. I find that recipients do not increase relative indirect costs after the gift and do not increase their relative asset reserves. At the 95% significance level, effects greater than 5% for indirect costs and roughly 20% for saving are ruled out.

Overhead ratio. The “overhead ratio,” defined as indirect costs divided by total spending, is the standard measure of indirect cost spending that donors use to evaluate nonprofits. Table 2 shows that the overhead ratio at recipient nonprofits neither increased nor responded differentially based on the percent of revenue that was restricted prior to the gift. In column (I), the baseline regression without the triple difference term, the coefficient β for the nonprofit’s overhead ratio is insignificant, so nonprofits did not increase their indirect spending after receiving Scott’s gift as a fraction of all spending. The mean value of the overhead ratio was 18% in 2019, meaning 18% of annual spending was devoted to indirect cost categories. The point estimate implies that the overhead ratio increased by 0.39pp, but this magnitude is economically small. The standard errors suggest that the test rejects, at the 95% confi-

dence level, effects that exceed a 0.9pp increase (or 5% increase from 2019) in the overhead ratio. Next, the full triple-difference specification of Equation (7) in column (II) shows that increasing the amount of restricted revenue for the nonprofit had prior to receipt did not differentially increase indirect spending. The term with an interaction between being treated and the percent of revenue restricted shows the higher treatment effect that restricted recipients could have had prior to the gift. The coefficient ϕ is insignificant, suggesting that a higher prior exposure to restrictions – a possible hallmark of underfunding – makes them no more likely to fund indirects. Figure A.8d shows Equation (7) (without the interaction term) separately by eight quantiles of the nonprofit’s annual revenues in 2019, comparing the treated units in that quantile of revenue versus their matched control nonprofits. As this analysis uses fewer treated observations in each quantile, noise increases. One could posit that smaller nonprofits may be more subject to nonprofit starvation cycle considerations. Confidence intervals suggest that overhead increases significantly for the smallest recipients, which would be worthy of further investigation with additional statistical power from additional post-period data.²¹

Taken together, through the lens of Proposition 2, the general lack of increase suggests that recipients were not constrained in their allocation to indirect costs prior to the gift. Importantly, if the aggregate test masks heterogeneity within size wherein some recipients increased overhead ratios and others decreased, then the hypothesis of Proposition 2 fails to hold, and the test cannot disprove that recipient nonprofits were constrained prior to the gift.

Expense to liquid asset ratio. Nonprofits, like consumers and firms, hold a reserve of liquid assets for emergencies or large changes in investment opportunities. The starvation cycle literature states that donors’ elasticity to the percentage of their gift used directly on charitable activities could drive suboptimal liquid savings. A large unrestricted donation should reduce that constraint and allow nonprofits to hold more buffer stock. I test relative spending out of total wealth and out of liquid wealth to create marginal propensity to consume (MPC) estimates analogous to estimates in Kaplan and Violante (2022). Table 2 shows that the gift did not change the average amount of buffer stock relative to annual spending for treated nonprofits by either definition. In column (III), the coefficient is insignificant on treatment for the expense to liquid asset ratio, defined as the sum of non interest bearing and interest bearing savings and investments on public markets. The confidence intervals imply that the reduction in expenses as a fraction of liquid assets was no larger than -5 or

²¹Additional robustness checks are shown in Figure A.8. Figure A.8a shows the regression repeated using Equation (5). It indicates that two years after gift receipt, the event time regression coefficient is significant but there is a potential for an increasing pre-trend in overhead ratios.

a 21% reduction in expenses as a fraction of assets compared to 2019. In column (IV), the interaction with percentage of revenue previously restricted did not create any differential effect. The lack of change implies recipients did not seem to have differentially larger liquidity constraints prior to the gift. This is robust to treating the nonprofit’s buffer stock as their full asset balance, as shown in columns (V) and (VI). In these units, the confidence intervals imply that the reduction in expenses as a fraction of assets was no larger than -0.30 or a 18% reduction in expenses as a fraction of assets. To investigate whether the starvation cycle could have held for just a subset of recipients, Figure A.8e runs Equation (7) (without the interaction term) separately by eight quantiles of the nonprofit’s annual revenues in 2019. The point estimates imply that savings could have increased meaningfully for the smallest quantile of nonprofits, but the effect is noisy. A potentially similar trend for the smallest nonprofits is visible in Figure A.8f but also statistically insignificant.

Taken together, there is not statistically significant evidence for liquidity constraints in treated nonprofits, even in subgroups of the nonprofit recipients. The null effects conflict with the prevailing hypothesis that nonprofits are close to financial distress from restricted donations and that, according to Proposition C.2, saving should increase if constrained. That said, the regression results describe only the average effect on the treated nonprofits, so as with the overhead ratio, heterogeneity between nonprofits is possible and would threaten the hypothesis of Proposition C.2.

Finding no significant average effect in overhead ratio and expense to asset ratios on treated nonprofits casts doubt on a starvation cycle experienced by windfall recipients. That said, the starvation cycle hypothesis could still be salient for untreated nonprofits, but importantly, information about Scott’s selection process makes the difference between the ATT and the ATE of a windfall ambiguous. On the one hand, careful selection on management could mean those recipients had already taken steps to overcome the starvation cycle by soliciting unrestricted funds and hence recipients were less constrained than the population. On the other, careful selection could mean chosen recipients were already the most aligned with donor incentives because they were the most constrained, and so the ATT could be larger than the average effect in the population.

6.3 Employee compensation

Figure 4 shows the change in the level and relative allocation of compensation of workers at recipient nonprofits. Figure 4b shows the estimate of τ_k from Equation (6) for total employee compensation. This estimate is the total wage bill at the nonprofit, including salaries and wages of non-senior and senior employees, plus other employee benefits, pension contributions and accruals, and payroll taxes. Recipients spent an increasing amount on

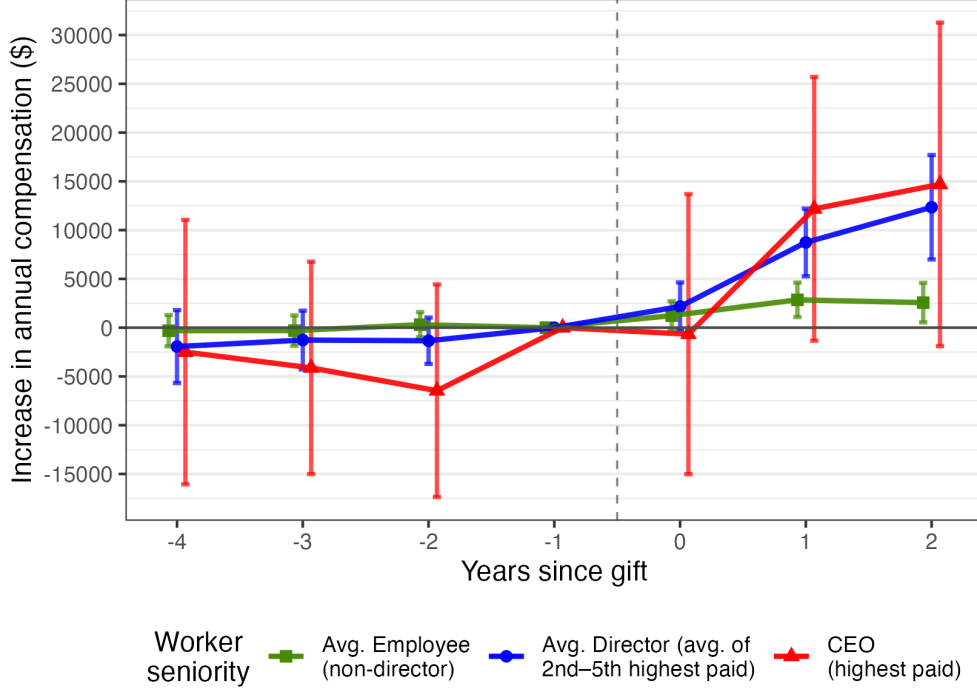
compensation after receiving a gift: \$0.02 per dollar of the average gift in the gift year, then \$0.09 and \$0.15 annually, comprising 26% of the gift principal two years later. Figure 4e shows the estimate of θ_k from Equation (5) for percent of spending allocated to employee compensation. Relative spending on compensation declined slightly as a counterweight to increased grant giving.

The increase in compensation stemmed from hiring and from higher pay for existing workers. Appendix Figure A.9 shows the estimate of θ_k from Equation (5) weighted by inverse pre-period revenue. Nonprofits hired 5.2 more employees two years after the gift – a 1.4% increase since a year prior to the gift. The remaining increase in compensation accrued through increasing compensation of workers. Figure 5 shows the increase in wages using θ_k from Equation (5) for three types of employees: the highest paid employee (my proxy for the chief executive officer), directors (the average wage of the second through fifth highest paid workers), and the average non-senior employee. This figure shows that compensation of the highest paid employee increased \$15K two years after the gift. The compensation increase for directors and for employees were \$12K and \$3K respectively. These estimates are increases off of different baselines; highest compensation increased 5.8% from the average in the year prior to the gift, versus 11.8% for directors and 5.5% for the average non-senior employee. (The relative increase is larger if using 2019 averages in Table 1 as a baseline.) Appendix Figure A.10 shows that the aggregate annual incidence of executive compensation of annual spending peaks at \$0.013 per dollar of the average gift two years after receipt. The increase in compensation for executives comes almost entirely from increases in base pay. Appendix Figure A.11 shows the same event study as Figure 5 but only for base pay (“salaries”) from recipients and control nonprofits whose Form 990 included Schedule J. Increases in salaries in this population are of almost identical magnitude to increases in bonus-inclusive total compensation.²²

Using senior wage increases as evidence of agency frictions in Proposition 3 requires invoking both Assumptions 2 and 3. In Table 3, I provide suggestive evidence of agency frictions with just Assumption 2 and one new restriction. Under the assumption that the marginal product of labor is equal for CEO’s who remained in the job after the gift versus newly hired CEO’s, but that the ability to allocate funds is greater for the pre-existing CEO,

²²Appendix Table A.1 shows that restrictedness of the target nonprofits did not yield significantly higher increases in director compensation. Indeed, at the 90% confidence level, more restricted nonprofits increased director compensation less. Appendix Figure A.12 and Appendix Table A.2 shows these estimates weighted by the inverse of nonprofit size – the weighting used for all other outcome variables in the paper up to this point. Unweighted is my preferred specification because employee wages do not exhibit the same size-related heteroskedasticity that the other spending-related outcomes do as discussed in more detail in Section 4.2. The estimate for the highest paid employees is quantitatively similar, but the estimated wage increase for non-senior employees is more modest than in the unweighted specification.

Figure 5: Wage increases by worker seniority



Note: This figure shows the incremental increase in total compensation at recipients for three types of employees: the chief executive officer, the average director, and the average non-senior employee relative to matched non-recipients. In red, the compensation of the chief executive (proxied by the highest paid employee at the nonprofit) increases \$15K more than compensation at matched controls two years after receiving a gift from Ms. Scott. In blue, the compensation of average director (proxied by the average compensation of the second through fifth highest paid employees) increased by an additional \$12K. In green, the average compensation of a non-senior employee increased by an additional \$3K.

compensation increases would have been smaller for new CEO's. To test for agency frictions, Table 3 shows the coefficients for CEO salaries in the following triple-difference regression

$$Y_{it}^c = \delta_t + \alpha_i + \sum_{k=-4, k \neq -1}^2 \tau_k^{CEO} 1(t - c = k, D_i = 1) + \phi_k(\text{New CEO})_i 1(t - c = k, D_i = 1) + \varepsilon_{it}. \quad (8)$$

The interpretation of the significant, negative coefficient ϕ^{CEO} is that nonprofits with CEO's installed in the year after receipt have smaller increases in salaries. The results are consistent with agency frictions. Strikingly, the point estimates imply that CEO salaries rose *only* for CEO's who remained in their position after their nonprofit received Scott's gift. In contrast,

nonprofits with new CEO’s in the post-period had CEO salaries that were essentially parallel with matched untreated organizations in the pre-period.

Formally extrapolating from the average effect on Scott’s recipients to the average treatment effect for any nonprofit is outside the scope of this paper. However, in the case of executive compensation, the ATT likely understates the treatment effect for an average nonprofit. The selection of recipients to have high governance, by way of Scott’s descriptions of “multi-year track records,” “measurement and evidence of outcomes”, and “experienced leadership” are all suggestive that results are a lower bound on executive pay increases after a windfall. Namely, if increases were due to agency frictions, such frictions would be more apparent in lower governance settings.

6.3.1 Comparing incidence with for-profits using present value

In this section, I compute what the point estimates imply for the present value of senior wages, in order to aid comparison with external executive compensation estimates. Unlike most other choices of input spending, wages exhibit downward nominal rigidity; nominal wages rise but rarely fall. As a result, the one-time increase in wages is more likely to be permanent than the increases in other spending categories such as grant giving or indirect costs. Figure 5 reports the estimates of τ_0 , τ_1 , and τ_2 . The translation of this estimate to the present value of those costs requires an assumption for the nonprofit’s discount rate r , as shown here:

$$PV_\tau = \tau_0 + \frac{\tau_1}{1+r} + \sum_{s=2}^{\infty} \frac{\tau_2}{(1+r)^s} \quad (9)$$

As a result with an assumption of the prevailing interest rate $r = 4.5\%$, the gift arriving one-time with once-and-for-all wage increases means that the top five executive wage increase actually comprised \$0.30 of every dollar of the average gift – nearly one third. Using the interest rate at the onset of Scott’s gift giving in 2020, $r = 0.10\%$, this estimate is far larger, exceeding the size of the gift.

A more conservative estimate also takes into account the contribution crowd in as part of windfall off of which nonprofits increased wages. Dividing the present value of wage increases by the present value of the gift and additional contributions yields an estimate of \$0.18 per dollar of windfall was allocated to wage increases for the top five executives.

[Ohrn \(2023\)](#) provides a recent estimate of spending out of cash windfalls for for-profit, publicly traded firms. His analysis of two tax changes suggests that \$0.17-0.25 of every windfall dollar is allocated toward the pay of the five highest paid employees. My analogous estimate, \$0.30, is larger but in the same range when adding contribution crowd in to the

Table 3: *This table shows the results of Equation (8) for CEO salaries, split by whether the CEO turned over in the year after the gift. Rows labeled “ $\times$ new CEO” report the differential effect for organizations where the highest-paid individual changed in the year after the gift (coefficients ϕ_k).*

	CEO salary (\$K)
$t = -4$	-1.15 (4.24)
$t = -3$	-5.18 (3.77)
$t = -2$	-1.19 (3.69)
$t = 0$	13.09*** (3.58)
$t = 1$	21.0*** (4.7)
$t = 2$	17.64*** (5.03)
$t = -4 \times (\text{new CEO})$	6.00 (9.75)
$t = -3 \times (\text{new CEO})$	9.00 (9.64)
$t = -2 \times (\text{new CEO})$	6.51 (7.63)
$t = 0 \times (\text{new CEO})$	-20.15* (8.19)
$t = 1 \times (\text{new CEO})$	-31.6** (11.9)
$t = 2 \times (\text{new CEO})$	-17.8 (12.3)
Num.Obs.	21632
R2	0.864

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

denominator (\$0.18). In sum, the windfall-related senior compensation increases at recipient nonprofits are quite similar to estimates seen at for-profit publicly traded firms.

6.3.2 Alternative mechanisms

The analysis above is most consistent with senior compensation increases stemming from agency frictions. Namely, that executives with control over residual cash flows have a greater ability to redirect unrestricted rather than restricted donations. However, the analysis also provides evidence *against* two other hypothesized mechanisms for employee compensation increases after cash windfalls: implicit compensation contracts and pay for performance.

First, wage increases due to an implicit compensation contract would take the form of employees temporarily lending funds to the firm on the condition of receiving bonuses in liquid times. [Howell and Brown \(2023\)](#) presents evidence of this mechanism in describing the wage increases at for-profit firms receiving research grants. Liquidity constrained firms, by definition, have a borrowing limit because of imperfect capital markets. As a result in this mechanism, firms find it easier to borrow from their employees. In that context, wage increases are not a function of effort but simply a contracted bonus written for firms with pre-existing liquidity constraints. The phenomenon predicts wage increases to scale with tenure at the firm, which is related to the seniority I find. Without seeing these contracts explicitly, I cannot entirely rule out this mechanism from consideration. However, this mechanism does necessitate pre-existing liquidity constraints at recipients and the gift relieving those constraints. In Table 2, I find scant evidence from relative choices to spend and save that recipients were liquidity constrained prior to the gift. Furthermore, if this mechanism were responsible, I would expect to see wage increases come primarily in the form of temporary bonuses in the year of the gift then diminish in the subsequent years. Instead, salaries increase consistently starting in the year of the gift.

Another explanation of the wage increases is pay for performance: that the raises fairly rewarded the increase in worker’s marginal product of labor. This channel could operate directly through rewarding workers for the receipt of the gift or for running a larger organization going forward. In the first case, the relative increase would have to be explained by recipients seeing the gift as a reward for past effort concentrated among senior employees. The compensation increases were larger for senior employees, so this mechanism would need the effort to have been expended on an activity done differentially by the senior employees, like fundraising or publicizing the organization ([Pagnoni, 2023](#)). However, pay for performance still does not address why the increase occurred in base pay rather than a one-time bonus timed with receipt of the gift. In this second case, production dramatically increased at nonprofits who received a gift. With the sizable increase in nonprofit spending, the marginal

product per CEO, director, and employee increased dramatically and workers could have rightfully been paid more (Gabaix and Landier, 2008). That said, this story is inconsistent with Table 3 because the story would predict equal base pay for new and incumbent CEO’s.

7 Conclusion

MacKenzie Scott’s gifts were massive in absolute and relative terms. They were also followed by additional follow on donations and grants. Nonprofits received 67% of the average gift in additional contributions and spent 95% of the gift within two years. While a large body of work has demonstrated such announcements of support can influence the additional funding nonprofits receive, much less is known about the ways that nonprofits would allocate these funds.

A model of nonprofits maximizing charitable output subject to donation restrictions guides the interpretation of this allocation. I document that the gift altered the allocation of recipient spending. In particular, nonprofits “on-granted” some of the windfall to other organizations and individuals. While in tax documents the social value of charitable output created is unobservable, under the assumption that nonprofit agent maximizes her objective, the change in allocation provided novel insight into the charitable production function. In particular, charitable production for recipients seemed to exhibit decreasing returns to scale. With nearly two million operating tax-exempt organizations in the United States, this finding may rationalize the fragmentation: a social planner facing diminishing marginal output at the entity level would optimally spread activity across many smaller entities rather than concentrate it among fewer larger ones.

As for evaluating the nonprofit starvation cycle, one potential interpretation of these findings is that internal stakeholders capture a portion of unrestricted funds. Executive compensation, frequently cited as a bellwether of the moral hazard problem that restrictions would mitigate, increases. The increases are similar in magnitude to estimates at for-profit firms and suggest that, in this way, nonprofits’ spending out of windfalls resembles that of for-profits. In contrast, the lack of significant changes in indirect cost spending and saving of liquid and illiquid assets suggests that recipients were not constrained in those allocations prior to the gift.

For nonprofit funders, the results are nuanced. While the analysis did not confirm the conjectured benefits of unrestricted gifts, it shows that unrestricted gifts are spent on charitable causes quickly, just as funders often hope. Moreover, unrestricted gifts could have other advantages over an otherwise equal restricted gift, such as a lower administrative burden, lighter monitoring, and deeper alignment with grantees. In light of such benefits, the

allocation of \$0.18 per dollar of a windfall, likely an upper bound on misallocation, could be modest compared to the costs of designing and enforcing donation restrictions.

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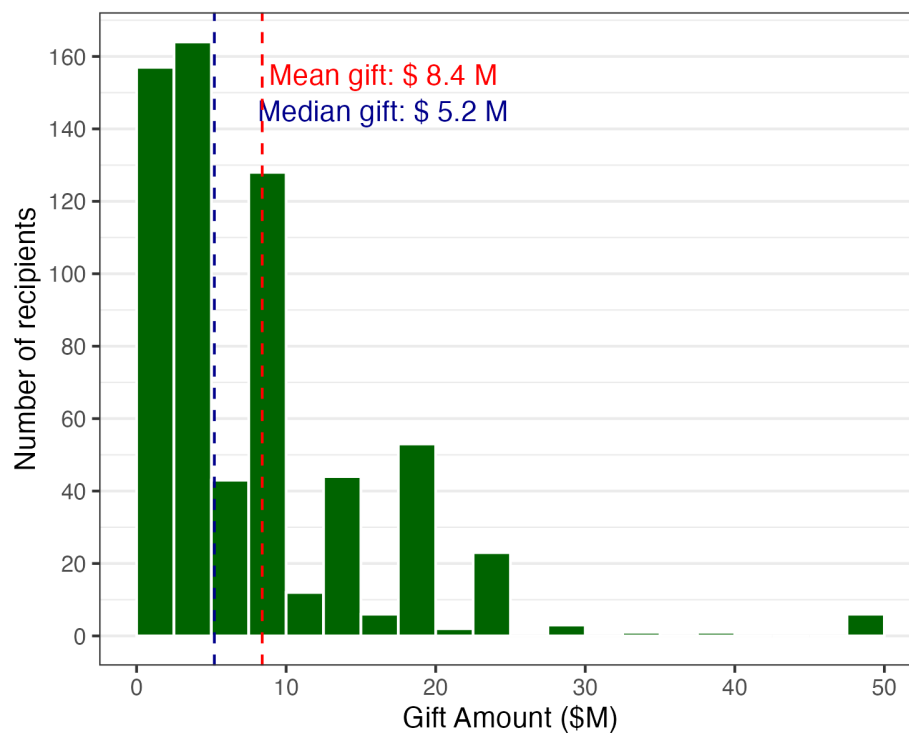
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A Appendix Tables and Figures

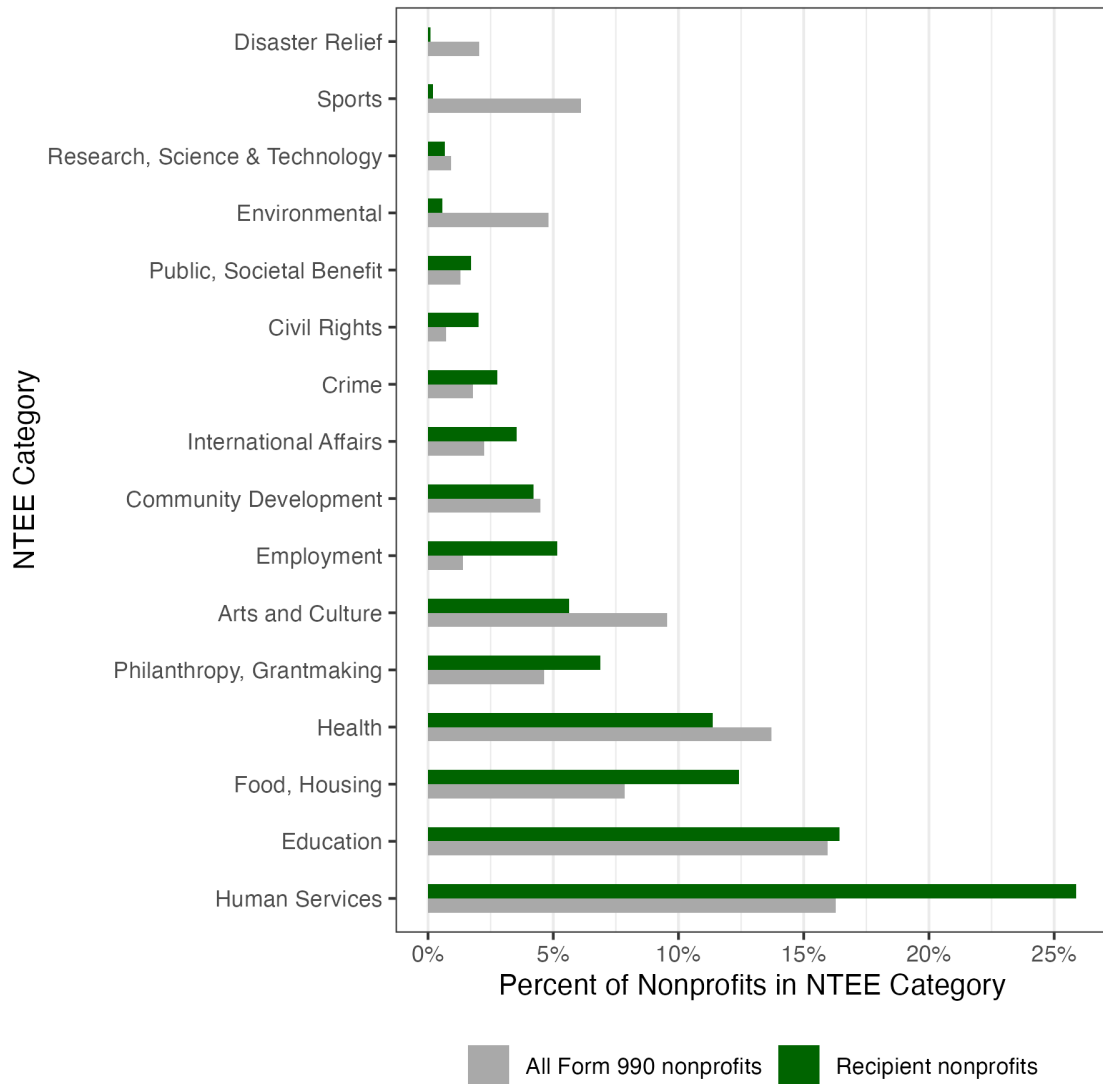
A.1 MacKenzie Scott's gifts: additional details

Figure A.1: *Distribution of gift amounts*



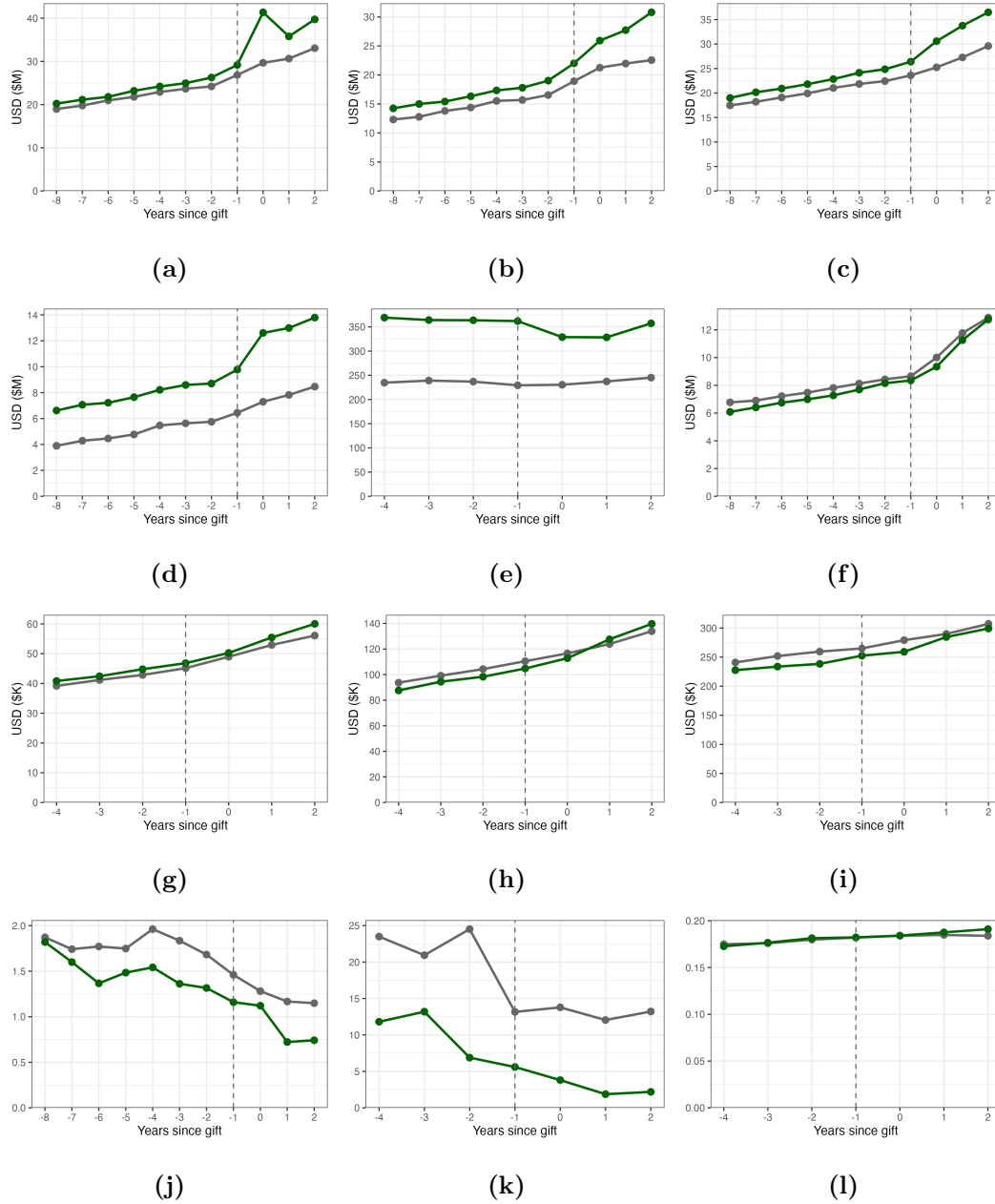
Note: This figure shows the distribution of MacKenzie Scott's gift amounts in millions of dollars.

Figure A.2: Sectoral distribution of gifts



Note: This figure shows the sectoral distribution of recipient nonprofits versus the full population of nonprofits who filed a Form 990 in 2019. On the Y-axis is a summary of the NTEE category: the first letter of the NTEE code for the organization. Codes C and D were combined into “Environmental.” Codes E, F, G, and H were combined into “Health.” Codes K and L were combined into “Food, Housing.” Codes O and P were combined into “Human Services.” Codes U and V were combined into “Research, Science & Technology.” The X-axis plots the percent of nonprofits in each category for two populations: all nonprofits filing a Form 990 and nonprofits in my sample of gift recipients.

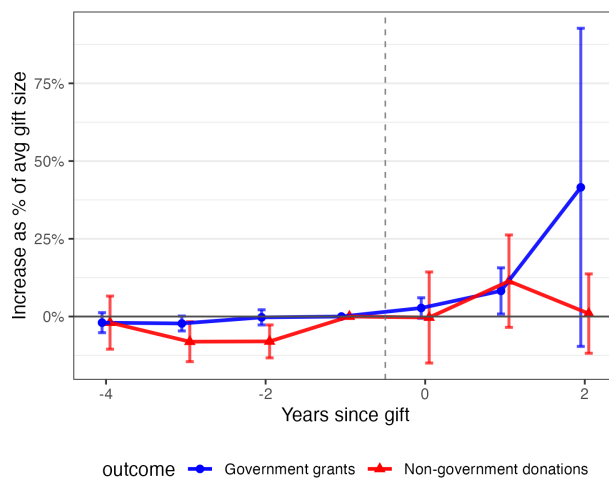
Figure A.3: Trends in key outcome variables for recipients and matched control nonprofits



Note: The panels plot the time series of annual mean outcomes for nonprofits who received an unrestricted gift from MacKenzie Scott (green) and nonprofits in the matched control (gray), relative to the gift year. (a) reports the dollars of revenue that the nonprofit received. (b) reports the dollars of donations and grants that the nonprofit received, in excess of MacKenzie Scott's gift, if treated, or total for the matched control. (c) plots the total expenses the nonprofit reported. (d) reports the dollars of grants given to other organizations and to individuals by the recipient nonprofit or control nonprofit. (e) plots the average number of employees reported on the nonprofit's W-3, (f) plots dollars allocated to compensation of employees. Compensation equals the sum of all salaries, benefits, and pension contributions of the nonprofit. (g)-(i) plot average compensation for all non-director employees, director employees and the highest paid employee. (j) reports the average expense to lagged asset ratio. (k) reports this ratio limiting to liquid assets only. (l) reports the percent of total annual spending devoted to "indirect cost" or "overhead" cost.

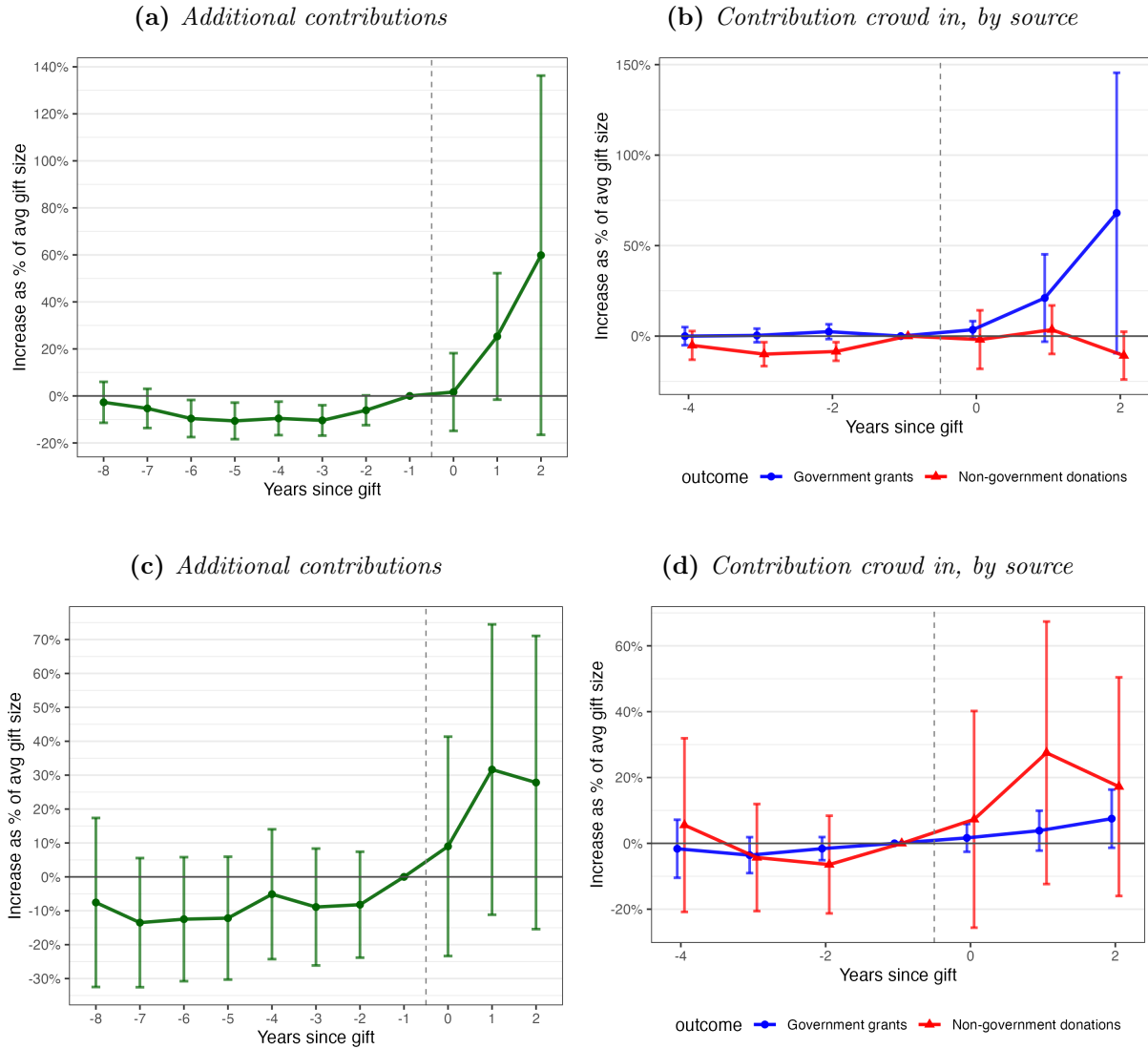
A.2 Contribution crowd in

Figure A.4: *Contribution crowd in, by source*



Note: This figure shows the breakdown of contribution crowd in, by the source of the contribution. The Y-axis shows the estimate of τ_k from Equation (6) for two outcomes: donations and government grants reported in addition to MacKenzie Scott's gift, measured as a fraction of the average gift, with 95% confidence intervals. The X-axis tracks years k since the gift was received. Estimates for the crowd in of government grants and additional donations are insignificant at the 95% confidence level.

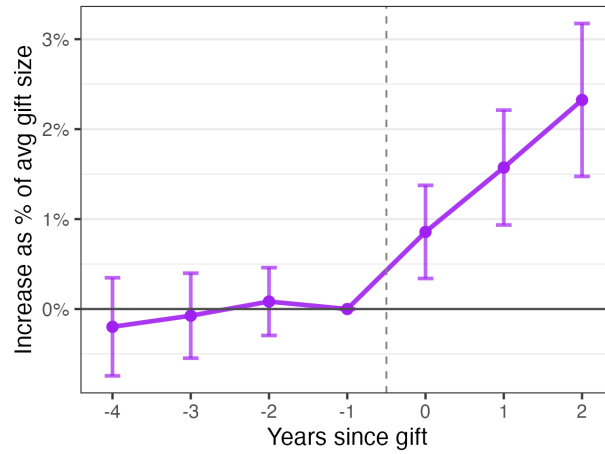
Figure A.5: *Crowd in by organization type*



Note: Crowd in at grantmakers (a, b) and direct service organizations (DSO's) (c, d) who received donations from MacKenzie Scott. Grant makers are defined as any recipient whose grants totaled more than 0.7% of their charitable expenses in 2019. Y-axis shows the estimate of τ_k from Equation (6) for total contributions (donations plus government grants) reported in addition to MacKenzie Scott's gift, measured as a fraction of the average gift, with 95% confidence intervals. The X-axis tracks years k since the gift was received. Estimates are noisy for both groups. However, point estimates suggest that crowd-in was driven more by government grants at grantmakers than at DSO's.

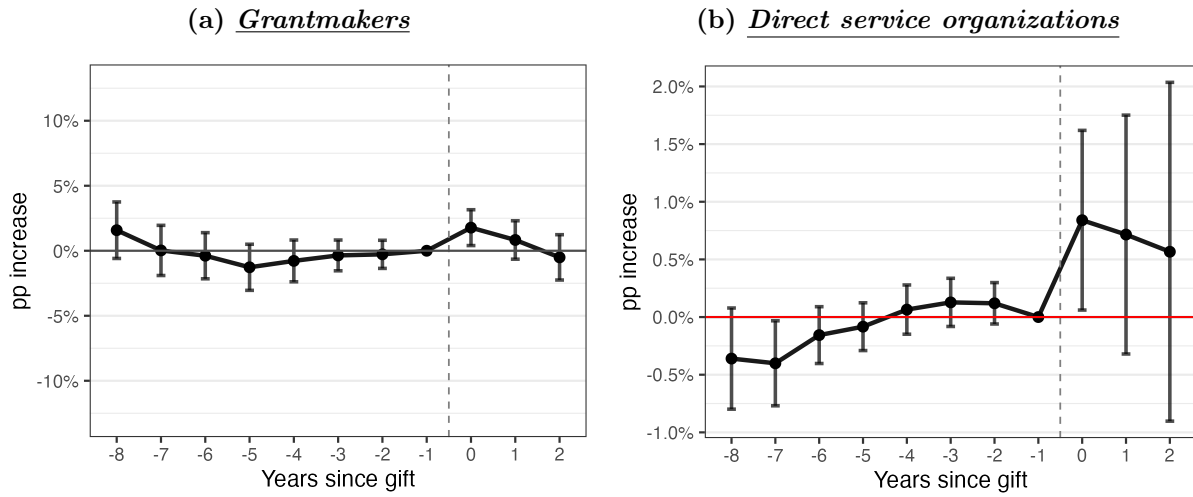
A.3 Allocation of spending

Figure A.6: *Fundraising expenses*



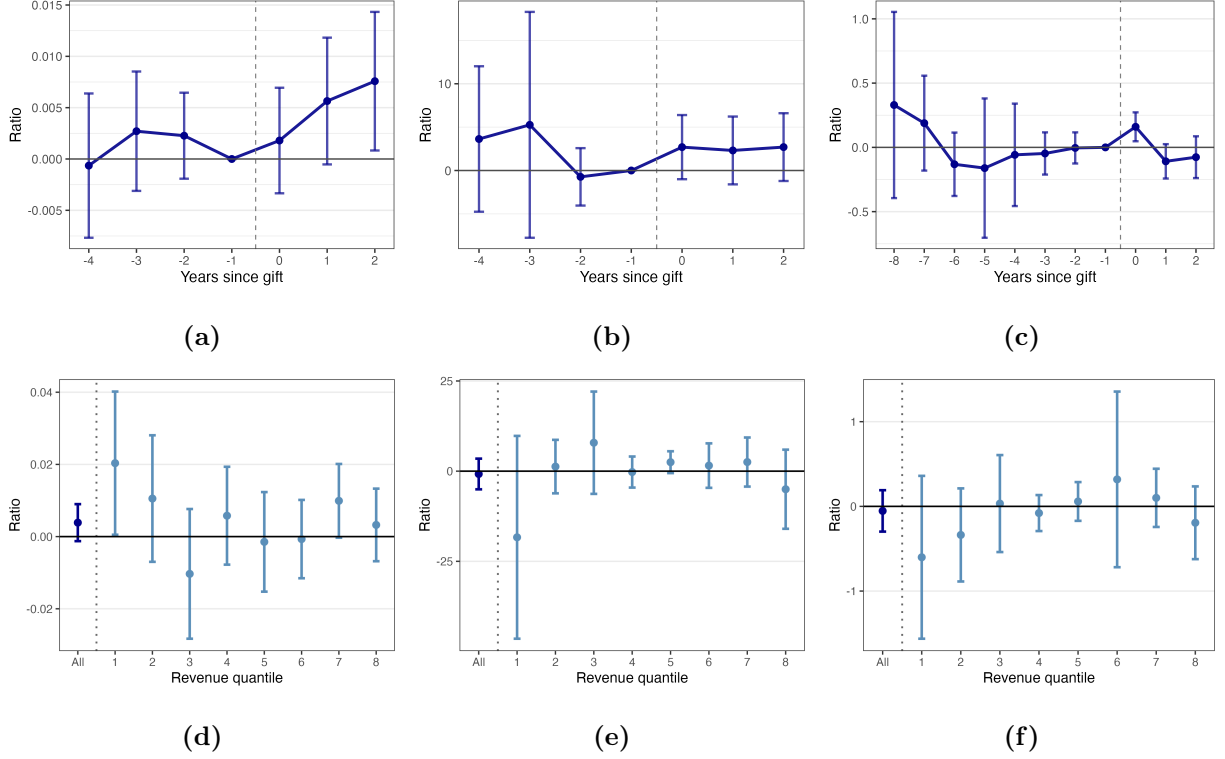
Note: This figure shows the increase in total fundraising expenses at recipient nonprofits compared to non-recipient nonprofits. Fundraising spending increased steadily to peak at 2% of the average gift size two years after the gift.

Figure A.7: Grant giving by organization type



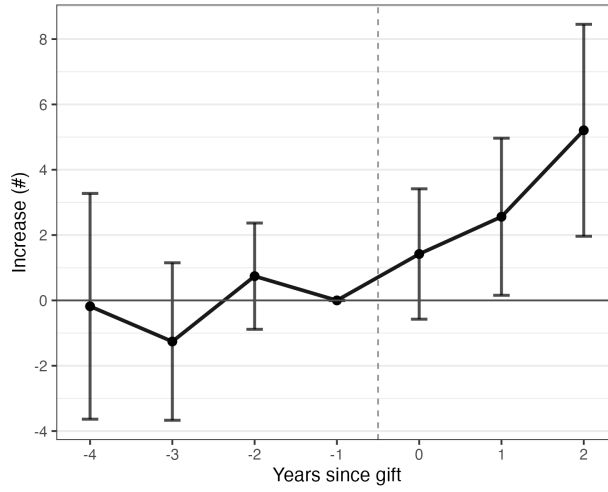
Note: (a) This figure shows the result of Equation (5) for percent of total expenses comprised of grants to other organizations. Grantmakers increased the percent of their annual expenses devoted to these grants at the 95% significance level in the year of the gift but not afterward. In the year prior to the gift, grants comprised 25.2% of annual spending for grantmakers. The Y-axis range is equal to the interquartile range of grants as a fraction of expenses for grant making organizations. (b) DSO's increased the percent of their expenses devoted to grants by 0.8 percentage points in the year of Ms. Scott's gift. In the year prior to the gift, grants comprised just 0.1% on average of annual spending. To demonstrate the size of this relative effect, red lines denote the interquartile range of grants as a fraction of expenses.

Figure A.8: *Robustness: Overhead and expense to asset ratios*



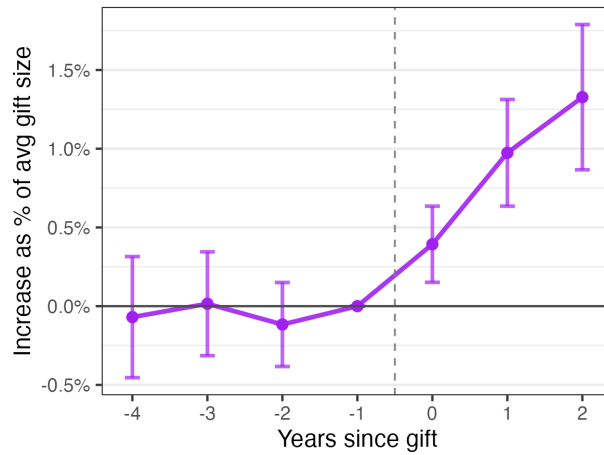
Note: The first row shows the estimates of θ_k from Equation (5) for three variables relevant to the nonprofit starvation cycle hypothesis. Error bars show 95% confidence intervals. Standard errors are clustered at the matching group level. Panel (a) plots the estimate for the overhead ratio. Panel (b) plots the estimate for the ratio of annual expenses as a fraction of the recipient's lagged liquid assets: defined as assets in checking accounts, savings accounts, and publicly traded securities. Panel (c) plots the same estimate using total assets. The second row shows the estimates of Equation (7) without the interaction term for percent of revenue that comes from restricted revenues. The dark blue estimate is from Table 2. Light blue show this estimate run separately for recipients versus their matched controls in by revenue quantile. Panel (d) plots the overhead ratio and shows that an estimate exceeding a 0.9pp (5%) increase in indirect cost spending as a fraction of total spending (the overhead ratio) is ruled out at the 95% confidence level. The point estimate is larger and significant for the smallest revenue quantile and smaller and insignificant for the largest revenue quantiles. Panel (e) plots the expense to lagged liquid asset ratio and shows that an estimate smaller than a -5 (21%) decrease is ruled out at the 95% confidence level. The point estimate for the smallest quantile is larger in magnitude than the larger quantiles. Panel (f) plots the same ratio using total assets and shows that an estimate smaller than a -0.30 (18%) decrease is ruled out at the 95% confidence level.

Figure A.9: Employees



Note: This figure shows the increase in the reported number of employees at recipients of Ms. Scott's gift. The Y-axis shows the estimate of θ_k for number of employees, with 95% confidence intervals, from Equation (5), weighted by inverse pre-period revenue. The X-axis tracks years k since the gift was received. Two years after the gift, the average nonprofit has 5.2 additional employees.

Figure A.10: Compensation of the five highest paid individuals



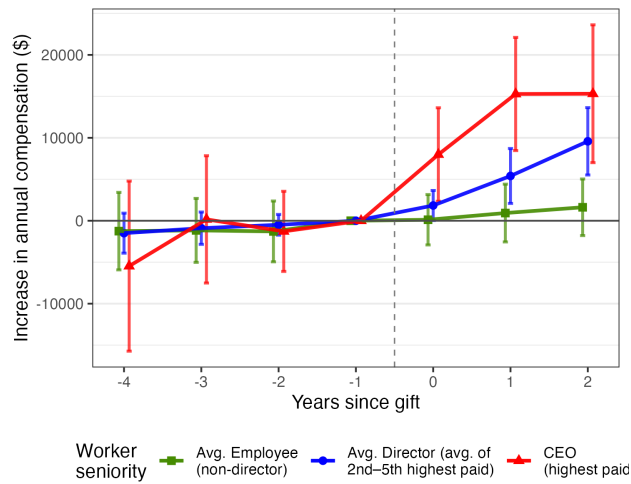
Note: This figure shows the increase in the sum of total compensation for the five highest paid at recipients, as a fraction of the average gift size. The Y-axis shows the estimate of θ_k , with 95% confidence intervals, from Equation (5), weighted by inverse pre-period revenue. Over two years since gift receipt, 2.7% of the average gift was spent on executive compensation. The X-axis tracks years k since the gift was received.

Figure A.11: *Base pay increases by worker seniority*



Note: This figure shows the increase in base pay at recipients for two types of employees: the chief executive officer and the average director. In red, the compensation of the chief executive (proxied by the highest paid employee at the nonprofit) increased \$15K more than compensation at matched controls two years after receiving a gift from Ms. Scott. In blue, the compensation of average director (proxied by the average compensation of the second through fifth highest paid individuals) increases \$13K. The underlying data derive from the Schedule J of the Form 990 and provide only partial coverage of the full analysis sample.

Figure A.12: *Wage increases by worker seniority (weighted)*



Note: This figure shows the increase in total compensation at recipients for three types of employees: the chief executive officer, the average director, and the average non-senior employee. In red, the compensation of the chief executive (proxied by the highest paid employee at the nonprofit) increases \$15K more than compensation at matched controls two years after receiving a gift from Ms. Scott. In blue, the compensation of average director (proxied by the average compensation of the second through fifth highest paid individuals) increases by an additional \$12K. In green, the average compensation of a non-senior employee increases by an additional \$3K. Observations are weighted by inverse of pre-period nonprofit size.

Appendix Table A.1: *This table shows the outcome of Equation (7) for number of employees and the average compensation of non-director employees, director employees, and the highest paid employee, where the treatment effect is pooled over 0-2 years after receiving the gift. Regressions for employee compensation, director compensation, and CEO compensation are unweighted, and the regression for number of employees is weighted by inverse pre-period revenue. First, recipients have three more employees (insignificant at the 95% significance level), less so if they were more restricted prior to receiving a gift (columns (I) and (II)). Average employee compensation at the treated nonprofits increased by \$2K at the 95% significance level (column (III)) without a significant impact of restrictedness (column (IV)). Director compensation increased by \$8K, but the differential effect of restrictedness is negative and insignificant. CEO compensation increases by on average \$11K after the gift at the 95% significance level, and \$12K for the completely unrestricted recipients at the 90% significance level (columns (VII) and (VIII)).*

	Number of employees		Employee compensation (\$K)		Director compensation (\$K)		CEO compensation (\$K)	
	(I)	(II)	(III)	(IV)	(V)	(VI)	(VII)	(VIII)
Post	2.99*	3.64	2.266**	2.30*	8.22***	11.52***	10.73*	12.16+
	(1.20)	(2.29)	(0.695)	(1.01)	(1.67)	(2.65)	(4.33)	(6.24)
Post × (Percent of rev. restricted)		-1.90		-0.109		-11.66+		-5.19
		(5.53)		(2.099)		(6.96)		(12.67)
Num.Obs.	21068	21061	18155	18148	20634	20627	21232	21225
R2	0.927	0.927	0.902	0.902	0.950	0.950	0.911	0.911

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Appendix Table A.2: *This table shows the outcome of Equation (7), for number of employees and the average compensation of non-director employees, director employees, and the highest paid employee, where the treatment effect is pooled over 0-2 years after receiving the gift. Observations are weighted by the inverse of pre-period revenue. First, recipients have three more employees (insignificant at the 95% significance level), more so if they were more restricted prior to receiving a gift (columns (I) and (II)). Average employee compensation at the treated nonprofits did not increase at the 90% significance level (column (III) and (IV)). Director compensation increased by \$6K at the 99% significance level, and the effect was significantly smaller for more restricted nonprofits (columns (V) and (VI)). CEO compensation increases by on average \$14K after the gift, and the effect was significantly smaller for more restricted nonprofits (columns (VII) and (VIII)).*

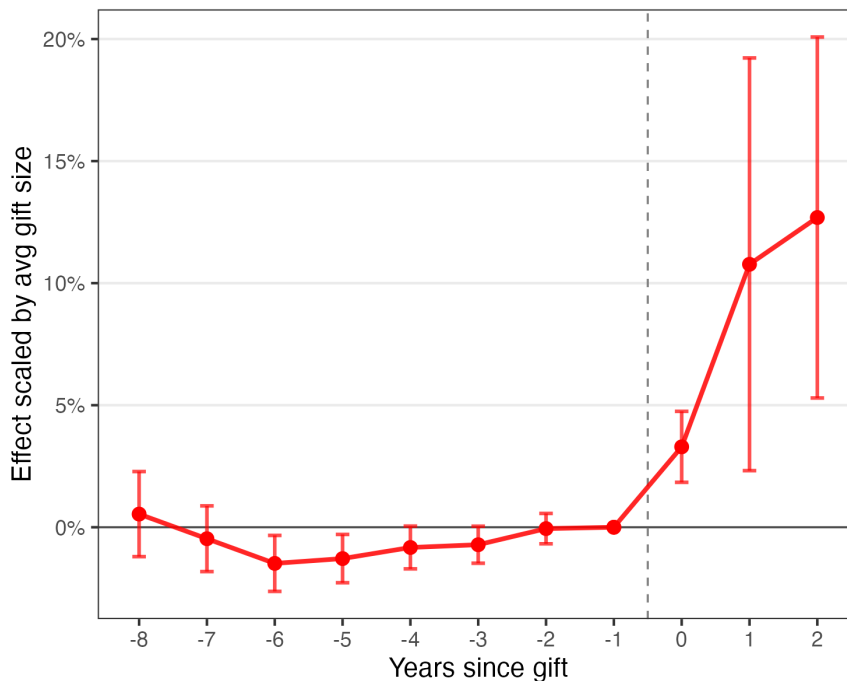
	Number of employees		Employee compensation (\$K)		Director compensation (\$K)		CEO compensation (\$K)	
	(I)	(II)	(III)	(IV)	(V)	(VI)	(VII)	(VIII)
Post	2.99*	3.64	1.65	0.00488	6.05***	12.72***	13.90***	23.10***
	(1.20)	(2.29)	(1.43)	(2.79026)	(1.34)	(2.49)	(3.16)	(5.78)
Post × (Percent of rev. restricted)		-1.90		4.75		-18.98***		-26.2+
		(5.53)		(5.26)		(5.38)		(14.7)
Num.Obs.	21068	21061	18155	18148	20634	20627	21232	21225
R2	0.927	0.927	0.860	0.860	0.837	0.837	0.836	0.836

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

B Estimating percent of average gift consumed

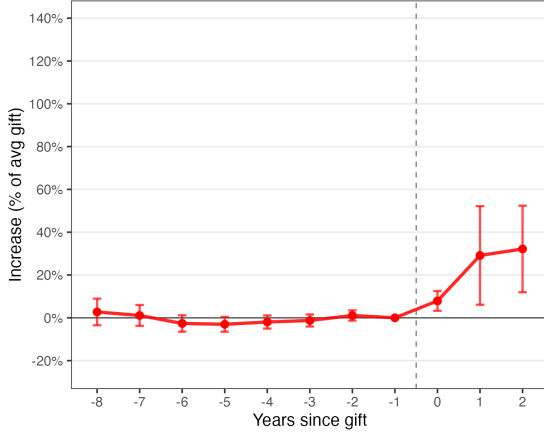
As discussed in Section 4.2, weighting while adding the average gift G_c into Equation (6) biases estimates of the percent of average gift consumed on outcome Y . Intuitively, the regression is weighted and G_c is not, and so G_c is “too large” due to larger nonprofits receiving larger gifts and the right skew of the gift size distribution. Without this procedure, Equation (6) produces Figure A.13 – estimates that are outside of the range of the four quartiles. As a result, I perform this regression separately for each revenue quartile of nonprofit. Figure A.14 plots each quartile’s regression coefficients separately and shows that computing their average estimate using the delta method produces a convex combination of the four estimates, shown in Figure 3.

Figure A.13: *Spending, without interacted regression*

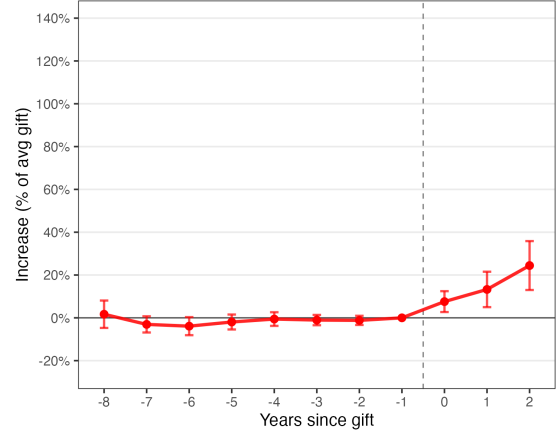


Note: This figure shows the result of estimating Equation (6) for the entire sample, without the interacted regression procedure of Golosov et al. (2024). Because of the right skewed distribution of gift amounts, the normalization is in essence “too large”.

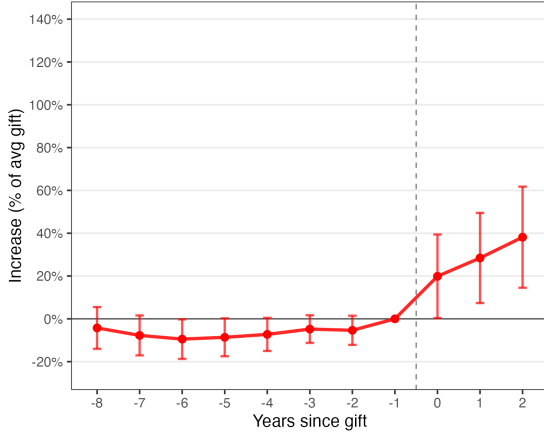
Figure A.14: *Spending by pre-period revenue quartile*



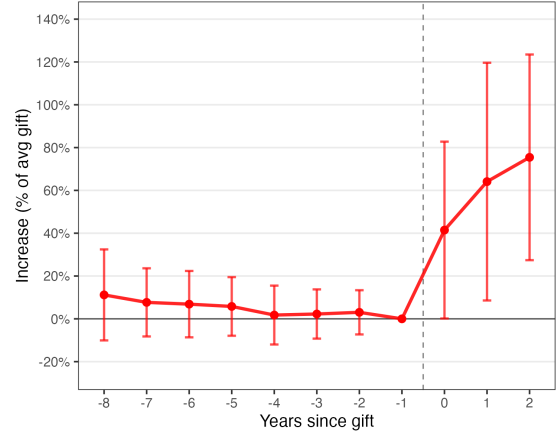
(a) *Quartile 1*



(b) *Quartile 2*



(c) *Quartile 3*



(d) *Quartile 4*

Note: This figure shows the estimate for the increase in expenses at recipient nonprofits, τ_k from Equation (6), as a fraction of average gift size, estimated separately for four quartiles of recipients by average pre-period revenue. Taking the average of these four estimates (by running a fully interacted regression and using the delta method) yields a more realistic answer than A.13.

C Proofs and additional model results

C.1 Proofs of Propositions

Proof of Proposition 1. This proof proceeds by enumerating how the non-negativity constraint μ for m could change from W to $W + G$ for every shape of $Y(\cdot)$. Define $W' \equiv W + G$.

1. Case 1: $Y(\mathbf{z})$ has decreasing returns to scale. Then we will invoke the results from Lemma 1, which provides a piecewise expression for $m^*(W)$ and $\mu^*(W)$ when $Y(\mathbf{z})$ has decreasing returns to scale. With these expressions, all that is left is to compute Δs for any possible $W \rightarrow W + G \equiv W'$.

For ease of notation, define the value function of donations, $\bar{\mathcal{D}}(W) \equiv \mathcal{D}(f^*(W), B(\mathbf{z}^*(W)))$.²³

- (a) $\mu(W) = \mu(W') = 0$. The gift moves the nonprofit from wealth W to wealth W' , and the parameters of the nonprofit's objective function are such that at both levels of wealth, $\mu = 0$. In Lemma 1, this is a case where $W, W' \geq e_0$. By definition of e_0 , $e_0 > 0$.

In this region, the change in the share of grant giving is

$$\Delta s = \frac{W' + \bar{\mathcal{D}}(W') - e_0}{W' + \mathcal{D}(W')} - \frac{W + \bar{\mathcal{D}}(W) - e_0}{W + \bar{\mathcal{D}}(W)} = \frac{-e_0}{W' + \bar{\mathcal{D}}(W')} - \frac{-e_0}{W + \bar{\mathcal{D}}(W)} > 0,$$

so long as $e_0 > 0$.

- In words, the non-negativity condition for m is slack before and after the gift, and $m^*(W) > 0$.
- (b) $\mu(W), \mu(W') > 0$.
 - The non-negativity constraint for m^* is binding before and after the gift, $m(W) = m(W') = 0$.
 - So, $s(W) = s(W') = 0$ and $\Delta s = 0$.
 - (c) $\mu(W) > 0, \mu(W') = 0$.
 - The non-negativity constraint is binding before the gift but slack afterward. This means that $m^*(W) = 0, m^*(W') = W' + \bar{\mathcal{D}}(W') - e_0$
 - So, $s(W) = 0, s(W') > 0$, and $\Delta s > 0$.
 - (d) $\mu(W) = 0, \mu(W') > 0$. This case is ruled out because $W' > W$.

2. Case 2: $Y(\cdot)$ has increasing returns to scale. The proof proceeds analogously to part 1 by invoking Lemma 2.

First, notice that for charitable production spending prior to the gift to be positive, as required in the proof assumption, $W \in [e_Z, \infty)$. That means that $W' \in (e_Z, \infty)$ as well. As a result, $m^*(W) = m^*(W') = 0$ and so $\Delta s = 0$.

²³The quantity B is defined in Proposition C.1 but the proof works for both this definition and the simpler definition $D(f^*(W))$ presented in the main text.

3. Case 3: $Y(\cdot)$ has constant returns to scale.

Note that for CRS to be true, $\lambda Y(\mathbf{z}) = Y(\lambda \mathbf{z})$. Then if we define $x \equiv \sum_k z^{(k)}$, then $Y(\mathbf{z}) = Y(x\boldsymbol{\xi}) = xY(\boldsymbol{\xi})$ where $\boldsymbol{\xi}$ is a vector where $\xi^{(k)} \equiv z^{(k)}/x$. So choosing the optimal $\boldsymbol{\xi}$ is choosing the optimal allocation of z 's into Y .

- (a) If $Y(\cdot)$ is such that $Y(\boldsymbol{\xi}) < b$, then an additional dollar spent on z 's is less valuable than an additional dollar spent on grants. The non-negativity constraints on grants is slack and the shares will remain the same after the gift. $m(W) = W$, so $\Delta s = 1 - 1 = 0$.
- (b) If $Y(\cdot)$ is such that $Y(\boldsymbol{\xi}) > b$, then the additional dollar spent on z 's is more valuable, so $\Delta s = 0 - 0 = 0$.

These choices are independent of the wealth of the nonprofit. They are functions of the production function and so the share of m doesn't change with the gift.

Putting these together, we conclude if $Y(\cdot)$ has non-decreasing returns to scale, then $\Delta s \leq 0$. This is the contrapositive of what we would like to prove. ■

Proof of Proposition 2.

We wish to compute

$$\Delta s_\iota(W) = s_\iota(W') - s_\iota(W)$$

where

$$s_\iota \equiv \frac{\iota^*(W)}{d^*(W) + \iota^*(W)}.$$

The principal's objective is:

$$\begin{aligned} p^*, d^*, \iota^*, m^*, f^* &= \arg \max_{p_i, d_i, \iota_i, m_i, f_i \geq 0} \theta U(p) + (1 - \theta) A_i d_i^\alpha \iota_i^{\beta_i} + (1 - \theta) b_i m_i \\ \text{such that } d_i + \iota_i + p_i + m_i + f_i &\leq W_i + \mathcal{D}(f_i) \quad [\text{Budget constraint}] \\ \text{such that } \iota_i &\leq W_i + (1 - \eta_i) \mathcal{D}(f_i) \quad [\text{Nonprofit starvation cycle (NSC) constraint}] \end{aligned} \tag{10}$$

Notice that fundraising can be solved offline and $D_i(f_i) - f_i$ can be replaced with an indirect donations function of optimal fundraising, defined as $\bar{D}_i$.

The new objective is:

$$\begin{aligned} p^*, d^*, \iota^*, m^* &= \arg \max_{p_i, d_i, \iota_i, m_i \geq 0} \theta_i U(p_i) + (1 - \theta) A_i d_i^\alpha \iota_i^{\beta_i} + (1 - \theta) b_i m_i \\ \text{such that } d_i + \iota_i + p_i + m_i &\leq W_i + \bar{D}_i \quad [\text{Budget constraint}] \\ \text{such that } \iota_i &\leq W_i + (1 - \eta_i) \bar{D}_i \quad [\text{Nonprofit starvation cycle (NSC) constraint}] \end{aligned} \tag{11}$$

Both the budget and NSC constraint distort the optimal choice. Note that the optimal choice of ι will be a typical Cobb-Douglas result if the NSC condition is slack.

- When the NSC constraint is slack,

$$s_\iota = \frac{\beta}{\alpha + \beta}. \quad (12)$$

because the optimal choice of indirect and direct costs will satisfy $\iota^* = \frac{\beta}{\alpha} d^*$.

- The constraint binding implies

$$s_\iota < \frac{\beta}{\alpha + \beta} \quad (13)$$

because the objective is increasing in ι .

To prove the implication, consider the effect of the gift G on s_ι by enumerating all possible cases of the bindingness of the NSC constraint before and after the gift. I omit i subscripts for clarity.

1. The NSC constraint is slack at W , $W + G$.

- $\Delta s_\iota = \frac{\beta}{\alpha + \beta} - \frac{\beta}{\alpha + \beta} = 0$.

2. The NSC constraint is binding at W and slack at $W + G$:

- Then $\Delta s_\iota > 0$. When the condition is slack at $W + G$, then $s_\iota(W + G) = \frac{\beta}{\alpha + \beta}$, and since the objective is increasing in ι , that means that $s_\iota(W) < \frac{\beta}{\alpha + \beta}$.

3. The NSC constraint is binding at W and $W + G$.

- Then $\Delta s_\iota \neq 0$ unless there is a knife-edge parameter restriction for η, α, β . I will show this assuming $\theta = 0$ then solving for Δs_ι explicitly to limit the complexity of the problem.
- There are two subcases, whether the non-negativity constraint on grants binds or not, that I will go through separately.

- (a) Non-negativity constraint for grants binds.

- Then the optimal choice of d^*, ι^* are:

$$\iota^* = W + (1 - \eta)\bar{D} \quad (14)$$

$$d^* = \eta\bar{D} \quad (15)$$

- And so

$$s_\iota = \frac{W + (1 - \eta)\bar{D}}{W + \bar{D}} = 1 - \eta \frac{\bar{D}}{W + \bar{D}} \quad (16)$$

- I will evaluate Δs_ι as a derivative $\frac{ds_\iota}{dG}$ in this case because the solution for s_ι will be differentiable in this region.

$$\frac{ds_\iota}{dG} = -\eta \frac{d}{dG} \left(\frac{\bar{D}}{W + \bar{D}} \right) \quad (17)$$

$$= \eta \frac{1}{W + \bar{D}} \left(\frac{\bar{D}}{W + \bar{D}} - \frac{W}{W + \bar{D}} \frac{d\bar{D}}{dG} \right) \quad (18)$$

Excluding a knife-edge parameter restriction $\frac{\bar{D}}{W + \bar{D}} = \frac{W}{W + \bar{D}} \frac{d\bar{D}}{dG}$, this expression will not equal zero for $\eta \in (0, 1]$.

(b) Non-negativity constraint for grants is slack.

- To solve for d^* explicitly, it means that

$$\iota^* = W + (1 - \eta)\bar{D} \quad (19)$$

$$d^* + m^* = \eta\bar{D} \quad (20)$$

Then using the first order condition that pins down m^* , we can solve for d^* explicitly:

$$b = A\alpha d^{\alpha-1} \iota^\beta \quad (21)$$

$$\Leftrightarrow b = A\alpha (d^*)^{\alpha-1} (W + (1 - \eta)\bar{D})^\beta \quad (22)$$

$$\Leftrightarrow (d^*)^{1-\alpha} = A\alpha b^{-1} (W + (1 - \eta)\bar{D})^\beta \quad (23)$$

$$\Leftrightarrow d^* = [A\alpha b^{-1} (W + (1 - \eta)\bar{D})^\beta]^{\frac{1}{1-\alpha}} \quad (24)$$

- This implies that the share devoted to overhead is:

$$s_\iota = \frac{W + (1 - \eta)\bar{D}}{W + d^*} = \frac{W + (1 - \eta)\bar{D}}{W + [A\alpha (W + (1 - \eta)\bar{D})^\beta]^{\frac{1}{1-\alpha}}} \quad (25)$$

- So the change in this share from a gift is:

$$\frac{ds_{\iota}}{dG} = \frac{d}{dG} \frac{W + (1 - \eta)\bar{D}}{K(G)} \quad (26)$$

$$\equiv \frac{E'(G)K(G) - E(G)K'(G)}{K^2(G)} \quad (27)$$

- Where

$$K(G) \equiv W + (A\alpha b^{-1})^{1/1-\alpha} E(G)^{\frac{\beta}{1-\alpha}} \quad (28)$$

$$K'(G) \equiv 1 + (A\alpha b^{-1})^{1/1-\alpha} \frac{\beta}{1-\alpha} E(G)^{\frac{\beta}{1-\alpha}-1} E'(G) \quad (29)$$

$$E(G) \equiv W + (1 - \eta)\bar{D} \quad (30)$$

$$E'(G) \equiv 1 + (1 - \eta) \frac{d\bar{D}}{dG} \quad (31)$$

So

$$\propto E'(G)K(G) - E(G) - (A\alpha)^{1/1-\alpha} \frac{\beta}{1-\alpha} E(G)^{\beta/1-\alpha} E'(G) \quad (32)$$

$$\propto -E(G) + E'(G) \left[K(G) - (A\alpha)^{1/1-\alpha} \frac{\beta}{1-\alpha} E(G)^{\beta/1-\alpha} \right] \quad (33)$$

$$\propto E'(G) \left[W + d^* - \frac{\beta}{1-\alpha} d^* \right] - (W + (1 - \eta)\bar{D}) \quad (34)$$

$$\propto (1 - (1 - \eta)) \frac{d\bar{D}}{dG} \left[W + d^* \frac{1 - \alpha - \beta}{1 - \alpha} \right] - \iota^* \quad (35)$$

where d^*, ι^* are functions of exogenous variables and written above.

- In general, the expression could be positive or negative, but it will be zero only in the event of a knife-edge parameter restriction. Two intuitive cases mean $\frac{ds_{\iota}}{dG} > 0$:
 - $\eta = 1$: All donations are restricted, and indirect cost funding can happen only from the nonprofit's existing endowment.
 - $\frac{d\bar{D}}{dG} = 0$: Donation crowd in is zero, so the gift does not cause an increase in unrestricted donations that allows the overhead ratio to increase.

Taken together, the solutions show that, with the exception of knife-edge parameter restrictions when the NSC constraint was binding before and after the gift, $\Delta s_{\iota} = 0$ implies that the NSC constraint was slack prior to the gift.

■

C.2 Additional results

C.2.1 Donation crowd in

How do donations respond after a wealth shock that coincided with an announcement of support? In order to distinguish factors that affect the donor's supply of donations from fundraising f_i , I introduce a parameterization for latent donor interest as marginal benefit of the donation. The explicit separation enables the inspection of mechanisms that could yield either crowd in or crowd out after a large public gift. For simplicity of notation, consider the case of a single charitable input ($K = 1$).

Assumption 4. *Nonprofit i at time t collects donations as a function of contemporaneous fundraising spending f_i and the donor's perception of the marginal charitable impact of funds B_i :*

$$D_i = \mathcal{D}(f_i, B_i)$$

where

$$B_i \equiv \mathbb{E}_{donor}[A_i \frac{\partial Y(z_i)}{\partial z_i}] \quad (36)$$

and $\mathcal{D}(\cdot)$ is differentiable and concave in both of its arguments.

With this assumption, the total derivative of donations D_i with respect to a gift from MacKenzie Scott is the sum of three terms:

$$\frac{dD_i}{dG_i} = \frac{\partial \mathcal{D}}{\partial f_i} \frac{df_i}{dG_i} + \frac{\partial \mathcal{D}}{\partial B_i} \left(\frac{d\mathbb{E}[A_i]}{dG_i} \frac{\partial Y(z_i)}{\partial z_i} + \mathbb{E}[A_i] \frac{\partial^2 Y(z_i)}{\partial z_i^2} \frac{dz_i}{dG_i} \right) \quad (37)$$

Equation (37) shows that donation crowd in captures a few effects simultaneously. First, the gift could increase fundraising spending. Second, the gift could increase the donor's perception of the marginal charitable impact of her own donation. Within this margin, the gift could either be a positive signal on the productivity of the nonprofit or could move the nonprofit up its production function either to increasing or decreasing returns to scale. While the empirical importance of the last two effects has been tested independently, the relative magnitudes have not. In the case that $\frac{dD_i}{dG_i} > 0$, rearranging and simplifying this expression yields Proposition C.1.

Proposition C.1. *Under Assumption 4, donation crowd in $\frac{dD_i}{dG_i}$ is positive if and only if*

$$\underbrace{-\frac{d \log \frac{\partial Y(z_i)}{\partial z_i}}{dG_i}}_{\substack{\text{decline in perceived value} \\ \text{of marginal spending} \\ \text{"economies of scale"}}} < \underbrace{\frac{d \log \mathbb{E}_d[A_i]}{dG_i}}_{\substack{\text{increase in} \\ \text{perceived quality} \\ \text{"announcement of} \\ \text{support"}}} + \underbrace{\frac{\varepsilon_f / \varepsilon_B}{\text{relative fundraising} \\ \text{vs. impact elasticity}}}_{\text{relative fundraising vs. impact elasticity}} \underbrace{\frac{d \log f_i}{dG_i}}_{\text{fundraising response}}$$

$$\varepsilon_f \equiv \left(\frac{f_i}{D_i} \frac{\partial D_i}{\partial f_i} \right) \quad \varepsilon_B \equiv \left(\frac{B_i}{D_i} \frac{\partial D_i}{\partial B_i} \right)$$

As Proposition C.1 shows, donation crowd in becomes a test of the relative magnitude of donors concerns about decreasing returns to scale and the positive signal that Scott's gift provided. In particular, donation crowd in implies that increased donations from fundraising and the increase in perceived productivity exceeded potential declining scale economies.

C.2.2 Saving

A dynamic nonprofit objective function must be used to examine the effect of constraints on the nonprofit's intertemporal saving problem, but the prior sections have assumed a one-period objective for simplicity. Assumption 5 provides the most incremental generalization required for analyzing savings behavior by allowing the nonprofit to allocate funds to charitable spending, grant giving, and fundraising over two periods.

Assumption 5 (The nonprofit starvation cycle hypothesis: savings).

The nonprofit agent chooses inputs z_{it} and z_{it+1} , grants m_{it} and m_{it+1} , private benefits p_{it} and p_{it+1} , fundraising f_i , and saving a_{it+1} to solve:

$$\begin{aligned}
 z^*, z_{t+1}^*, m^*, m_{t+1}^*, p^*, p_{t+1}^*, f^* = & \arg \max_{z_{it}, z_{t+1}, m_i, m_{it+1}, p_{it}, p_{it+1}, f_i, \geq 0} \left\{ (1 - \theta_i) [A_i Y(z_i) + b_i m_i] + \theta_i U(p_i) \right. \\
 & \left. + \beta (1 - \theta) [A_i Y(z_{it+1}) + b_i m_{it+1}] + \beta \theta_i U(p_{it+1}) \right\}, \\
 \text{s.t. } & W_i + D_i(f_{it}) - f_{it} = \sum_k z_{it}^{(k)} + m_{it} + p_{it} + a_{it+1} \quad [\text{Budget constraint 1}] \\
 \text{s.t. } & Ra_{it+1} = \sum_k z_{it+1}^{(k)} + m_{it+1} + p_{it+1} \quad [\text{Budget constraint 2}] \\
 \text{and } & a_{it+1} \leq W_{it} + (1 - \eta_i) D_i(f_i) \quad [\text{Nonprofit Starvation Cycle constraint}]
 \end{aligned}$$

for fraction $\eta_i > 0$ of donations that cannot be used for saving, $Y(z_i) \equiv \prod_{k=1}^K z_{ik}^{\phi_k}$, discount rate β , rate of return on savings R , and $\beta R < 1$.

Proposition C.2. Define the spending share,

$$s_a(W) \equiv \frac{W + D(f^*(W)) - a_{t+1}^*(W)}{a_{t+1}^*(W)}.$$

Under Assumption 5, $\theta = 0$, and no grant giving at either W or $W + G$ ($m^*(W) = m^*(W + G) = 0$), then if the change in the spending share after the gift,

$$\Delta s_a \equiv s_a(W + G) - s_a(W) = 0,$$

then “nonprofit starvation cycle” constraint was slack prior to the gift.

The proof is similar to that of Proposition 2, but the constraint now affects the choice of saving from the first and second period, a_{it+1} . When the NSC constraint is slack and under a general form of $Y(\cdot)$, the spending share will vary with wealth. A change in the savings share reveals either that the recipient faced a binding NSC constraint or that the recipient had a slack NSC constraint but not Cobb-Douglas preferences.

Proof of Proposition C.2. The logic follows Proposition 2. We will again omit i subscripts for clarity. The main difference is that this proposition builds off a dynamic model. Because $\beta R < 1$, a nonprofit choosing to allocate spending toward grants will only do so in the first period, not the second. Also, this Proposition further requires being in the regime where grant giving is not optimal, namely $m^*(W) = m^*(W + G) = 0$. Following Proposition 2, compute $d^*, \iota^*, d_{t+1}^*, \iota_{t+1}^*$, and therefore s_a , based on whether the NSC constraint is slack or binding. Again, choice $f^*(W)$ can be solved offline, and $D_i(f^*(W)) - f^*(W)$ can be replaced with $\bar{D}_i$. The vector of inputs is direct and indirect costs, $z_i = [d_i, \iota_i]$ but this can be generalized.

- Not binding NSC:

The first order conditions are:

$$\theta U'(p^*) = \lambda \tag{38}$$

$$\theta \beta R U'(p_{t+1}^*) = \lambda \tag{39}$$

$$a_{t+1}^* + d^* + \iota^* + p^* = W + \bar{D} \tag{40}$$

$$R a_{t+1}^* = d_{t+1}^* + \iota_{t+1}^* + p_{t+1}^* \tag{41}$$

Then the first order conditions for d^*, ι^* imply that they are in constant ratios with each other and across periods:

$$(\iota_t^*)^\beta (d_t^*)^{\alpha-1} = \beta R \iota_{t+1}^\beta (d_{t+1}^*)^{\alpha-1} \tag{42}$$

$$\Rightarrow \frac{d_{t+1}^*}{d^*} = \frac{\iota_{t+1}^*}{\iota^*} = (R\beta)^{\frac{1}{1-\alpha-\beta}} \tag{43}$$

$$s_a \equiv \frac{d^* + \iota^* + p^*}{a_{t+1}^*} = \frac{d^* + \iota^* + p^*}{\beta^{\frac{1}{1-\alpha-\beta}} R^{\frac{1}{1-\alpha-\beta}-1} (d^* + \iota^*) + p_{t+1}^*} \quad (44)$$

Notice that the derivative of the spending share with respect to the gift is non-zero when it is slack. But, one case where it is zero is when $\theta = 0$, so $p_t^* = p_{t+1}^* = 0$ and so the ratio is constant as a function of wealth.

- Binding NSC:

$$a_{t+1}^* + \iota_t^* = W + (1 - \eta)\bar{D} \quad (45)$$

$$d^* + \iota^* = \eta\bar{D} \Rightarrow \frac{1 + \beta}{\alpha} \iota^* = \eta\bar{D} \quad (46)$$

Then

$$\frac{d^* + \iota^* + p^*}{a_{t+1}^*} = \frac{\eta\bar{D}}{W + (1 - \eta)\bar{D} - \frac{\eta\alpha}{1+\beta}\bar{D}} \quad (47)$$

$$= \frac{\eta\bar{D}}{W + (1 - \eta(1 + \frac{\alpha}{1+\beta}))\bar{D}} \quad (48)$$

With this established, a wealth shock W to $W + G$ can be examined by enumerating the possible cases.

1. When the NSC constraint is slack at $W, W + G$ then $\Delta s_a \neq 0$. $\Delta s_a = 0$ under $\theta = 0$.
2. When the NSC constraint is binding at W and slack at $W + G$, then $\Delta s_a > 0$ as an implication of the constraint operating as a maximum value of a_{t+1} .
3. When the NSC constraint is binding at W and $W + G$, then $\Delta s_a \neq 0$.

- Notice that

$$\frac{d}{dG} \frac{d^* + \iota^* + p^*}{a_{t+1}^*} = \eta \frac{d}{dG} \frac{\bar{D}}{W + (1 - \eta(1 + \frac{\alpha}{1+\beta}))\bar{D}} \quad (49)$$

- Which in general is not equal to zero.

As a result, $\Delta s_a = 0$ implies that the NSC constraint was slack prior to and after the gift G .

■

C.3 Lemmas

Lemma 1. *Define*

$$e_0 \equiv \tilde{f} + \tilde{p} + c(\tilde{\mathbf{z}})$$

where the choice variables are the solution to

$$\begin{aligned} \tilde{f}, \tilde{p}, \tilde{\mathbf{z}}, \tilde{m} \equiv \arg \max_{\mathbf{z}_i \geq 0, p_i \geq 0, f_i \geq 0, m_i} \left\{ (1 - \theta_i) [A_i Y(\mathbf{z}_i) + b_i m_i] + \theta_i U(p_i) \right\}, \\ \text{s.t. } W + \mathcal{D}(f_i, B(\mathbf{z}_i)) - f_i = c(\mathbf{z}_i) + m_i + p_i. \end{aligned} \quad (50)$$

where $\tilde{\mathbf{z}}$ is a vector of inputs to Y and $c(\tilde{\mathbf{z}}) \equiv \sum_{k=1}^K z_k$.

Define $m^*(W)$ as the value m^* solving the maximization:

$$\begin{aligned} f^*, p^*, \mathbf{z}^*, m^* \equiv \arg \max_{\mathbf{z}_i \geq 0, p_i \geq 0, f_i \geq 0, m_i \geq 0} \left\{ (1 - \theta_i) [A_i Y(\mathbf{z}_i) + b_i m_i] + \theta_i U(p_i) \right\}, \\ \text{s.t. } W + \mathcal{D}(f_i, B(\mathbf{z}_i)) - f_i = c(\mathbf{z}_i) + m_i + p_i \end{aligned} \quad (51)$$

Let $\mu(W)$ be the Lagrange multiplier in the complementary slackness condition for the choice of $m^*(W)$.

Then for $Y(z)$ with decreasing returns to scale:

$$m^*(W) = \begin{cases} 0 & W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) \leq e_0 \\ W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) - e_0 & W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) > e_0 \end{cases} \quad (52)$$

$$\mu(W) = \begin{cases} \mu' \in \mathbb{R}_{\geq 0} & W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) \leq e_0 \\ 0 & W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) > e_0 \end{cases} \quad (53)$$

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Proof. Equation (50) is identical to Equation (51) except that it does not impose the non-negativity constraint for grants m .

In Equation (50), the FOC's are

$$1 = \mathcal{D}_f(\tilde{f}, B(\tilde{\mathbf{z}})) \quad (54)$$

$$\lambda(W) + \frac{\partial B}{\partial z_k} \mathcal{D}_B(\tilde{f}, B(\tilde{\mathbf{z}})) = (1 - \theta) A_i Y_{z_k}(\tilde{\mathbf{z}}) \quad \forall k \quad (55)$$

$$\lambda(W) = \theta U'(\tilde{p}) \quad (56)$$

$$\lambda(W) = (1 - \theta)b \quad (57)$$

$$\tilde{m} = W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) - (\tilde{f} + c(\tilde{\mathbf{z}}) + \tilde{p}) \Leftrightarrow \tilde{m} = W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) - e_0 \quad (58)$$

²⁴Both Lemma 1 and Lemma 2 use the augmented definition of the donation production function introduced in Assumption 4. Using $D_i = D(f_i)$ or $D_i = \mathcal{D}(f_i, B_i)$ does not change the result.

I represent the Lagrange multipliers as functions of W , $\lambda(W)$, to clarify how their values change with the exogenous state variable W that will change with Scott's gift.

The FOC's for each choice variable in (51) are:

$$1 = \mathcal{D}_f(f^*, B(\mathbf{z}^*)) \quad (59)$$

$$\lambda(W) + \frac{\partial B}{\partial z_k} \mathcal{D}_B(\tilde{f}, B(\tilde{\mathbf{z}})) = (1 - \theta)A_i Y_{z_k}(\mathbf{z}^*) \quad \forall k \quad (60)$$

$$\lambda(W) = \theta U'(p^*) \quad (61)$$

$$\lambda(W) - \mu(W) = (1 - \theta)b \quad (62)$$

$$m^* = W + \mathcal{D}(f^*, B(\mathbf{z}^*)) - (f^* + c(\mathbf{z}^*) + p^*) \quad (63)$$

In Equation (51) with a non-negativity constraint for grants m , there is also a complementary slackness condition,

$$\mu(W) \cdot m^* \geq 0. \quad (64)$$

This condition means that for a given exogenous wealth W , $\mu(W) = 0$ iff $m^* > 0$ and $\mu > 0$ iff $m^* = 0$.

Notice that if the non-negativity condition for m is slack ($\mu(W) = 0$), then the two sets of first order conditions are identical. In this region:

$$\begin{aligned} \tilde{f} &= f^* \\ \tilde{p} &= p^* \\ c(\tilde{\mathbf{z}}) &= c(\mathbf{z}^*) \end{aligned}$$

Because of (57), both e_0 and $\mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}}))$ are invariant to wealth. The choice variables are pinned down with a Lagrange multiplier $\lambda(W)$ that remains equal to $(1 - \theta)b$ regardless of the nonprofit's endowment W . So, using the complementary slackness condition for the first row and Equation (63) for the second, m^* has the following form:

$$m^*(W) = \begin{cases} 0 & \mu(W) > 0 \\ W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) - e_0 & \mu(W) = 0 \end{cases} \quad (65)$$

The goal, however, is to show $m^*(W)$ in cases explicitly as a function of W . Using the if and only if logic in the complementary slackness condition again,

$$m^*(W) = \begin{cases} 0 & W \leq e_0 - \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) \\ W + \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) - e_0 & W > e_0 - \mathcal{D}(\tilde{f}, B(\tilde{\mathbf{z}})) \end{cases} \quad (66)$$

This proves the lemma. Intuitively, there is a threshold after which the marginal value of grant giving outweighs the marginal value of charitable production using inputs $\mathbf{z}$, fundraising, and private benefits. Before wealth plus donations reaches that threshold, $m^*(W) = 0$. After reaching this threshold, the remaining funds will go to grant giving.

■

Lemma 2. Suppose $Y(z)$ has increasing returns to scale in the maximization problem (1). Define

$$\tilde{f} \equiv \{f | 1 = \mathcal{D}_f(f, B(0))\} \quad (67)$$

$$\tilde{p} \equiv \{p | \theta U'(p) = (1 - \theta)b\} \quad (68)$$

$$e_P \equiv \tilde{f} + \tilde{p} \quad (69)$$

$$e_Z \equiv \{\hat{f} + c(\hat{z}) | b + \frac{\mathcal{D}_B(f, B(\hat{z}))}{1 - \theta} = A_i Y_{z_k}(\hat{z}) \quad \forall k, \quad 1 = \mathcal{D}_f(f, B(\hat{z}))\} \quad (70)$$

Then for every $W \geq 0$, the optimal $m^*(W)$ is given as follows:

1. **If** $\exists k$ **s.t.** $b < A_i Y_{z_k}(0)$.

$$m^*(W) = 0, \quad \mu(W) > 0, \quad \forall W > 0$$

2. **If** $b \geq A_i Y_{z_k}(0) \forall k$.

When $\theta U'(0) < b(1 - \theta)$

$$m^*(W) = \begin{cases} W + \mathcal{D}(\tilde{f}, B(0)) - \tilde{f} & W + \mathcal{D}(\tilde{f}, B(0)) \in [0, e_Z) \\ 0, & W + \mathcal{D}(\tilde{f}, B(0)) \in [e_Z, \infty) \end{cases} \quad (71)$$

When $\theta U'(0) \geq b(1 - \theta)$ and $e_P < e_Z$

$$m^*(W) = \begin{cases} 0, & W + \mathcal{D}(\tilde{f}, B(0)) \in [0, e_P) \\ W + \mathcal{D}(\tilde{f}, B(0)) - e_P, & W + \mathcal{D}(\tilde{f}, B(0)) \in [e_P, e_Z) \\ 0, & W + \mathcal{D}(\tilde{f}, B(0)) \in [e_Z, \infty) \end{cases} \quad (72)$$

When $\theta U'(0) \geq b(1 - \theta)$ and $e_P \geq e_Z$

$$m^*(W) = 0, \quad \mu(W) > 0, \quad \forall W > 0 \quad (73)$$

Proof. We split into two primary cases based on the comparison of b and $A_i Y_{z_k}(0)$.

1. Case 1: $\exists k$ s.t. $b < A_i Y_{z_k}(0) \Leftrightarrow$ grants are never given.

In words, even at $x = 0$ the marginal payoff $A_i Y_{z_k}(0)$ exceeds b .

From here, since $Y(z)$ has increasing returns to scale, the objective will grow without bound.

This means that in this region, $m = 0$ because funds will be allocated to z .

Hence

$$m^*(W) = 0, \quad \text{for all } W$$

2. Case 2: $b \geq A_i Y_{z_k}(0) \quad \forall z_k \Leftrightarrow$ grant giving is valuable at low levels of wealth.

First, define $\tilde{f}$ as the principal's choice of fundraising expenditure when charitable production is zero. I.e., $\tilde{f}$ where $1 = \mathcal{D}_f(\tilde{f}, B(0))$.

- (a) **Condition** $\theta U'(0) < (1 - \theta)b$

Define hat versions of the choice variables as:

$$b = A_i Y_{z_k}(\hat{\mathbf{z}}) \quad \forall k. \tag{74}$$

and the fundraising first order condition holds

$$1 = \mathcal{D}_f(\hat{f}, B(\hat{\mathbf{z}})). \tag{75}$$

Then define $e_Z \equiv \hat{f} + \hat{p} + c(\hat{\mathbf{z}})$.

At this wealth + donations level, $\hat{p} = 0$ because $\theta U'(0) < (1 - \theta)b$ and $U'(p)$ is decreasing in p because $U(p)$ is a concave function. e_Z is independent of wealth because $\hat{\mathbf{z}}$ is chosen as $b = A_i Y_{z_k}(\mathbf{z}) \forall k$ and hence independent of wealth. This implies $\tilde{f}$ is independent of wealth.

- i. **Domain** $W + \mathcal{D}_f(\tilde{f}, B(0)) < e_Z$ Because Y has increasing returns to scale, the optimal choice of z is zero.

This leaves the remaining wealth to be allocated to f^* and m^* according to the first order conditions:

$$1 = \mathcal{D}_f(\tilde{f}(W), B(0)) \tag{76}$$

$$m^*(W) = W + \mathcal{D}(\tilde{f}(W), B(0)) - \tilde{f} - 0 - 0 \tag{77}$$

As a result, the principal chooses the optimal f keeping in mind zero spending on x and then can only allocate the remaining $W + \mathcal{D}(\tilde{f}, B(0))$ to m .

- ii. **Domain** $W + \mathcal{D}_f(\tilde{f}, B(0)) > e_Z$. Then $m^* = 0$ as marginal benefit loads on z^* .

The result is:

$$m(W) = \begin{cases} W + \mathcal{D}(\tilde{f}, B(0)) - \tilde{f} & [0, e_Z) \\ 0 & [e_Z, \infty) \end{cases} \quad (78)$$

(b) **Condition** $\theta U'(0) > (1 - \theta)b$. With this parameter assumption, there is a region where wealth is allocated to private benefits until the marginal utility of private benefits equals the marginal utility of grants, but the derivation is otherwise analogous.

- i. **Domain** $W + \mathcal{D}(\tilde{f}, B(0)) < e_Z$ In this domain, because $Y_{z_i}(z)$ is non-decreasing, the optimal choice of z^* is zero.

Fundraising f , like in (a), will be set to $\tilde{f}$. This will not vary with wealth because the second argument $B(0)$ is constant until a large enough wealth when B changes.

Since $U(\cdot)$ is strictly concave and $\theta U'(0) > b(1 - \theta)$, e_P will be a unique threshold/single crossing.

Define $e_P \equiv \tilde{f} + \tilde{p}$ as the wealth + donations level where the marginal value of grant private benefits satisfy their first order conditions while $z = 0$:

$$\tilde{p} \equiv \{p | (1 - \theta)b = U'(p)\}, \quad \text{and} \quad \tilde{f} = \{f | 1 = \mathcal{D}_f(\tilde{f}, B(0))\}$$

However, allocating to p will only be optimal if the marginal benefit of allocating a marginal dollar to x are not higher. In other words, if wealth is in the domain less than e_Z . So, there are two cases:

A. Case A: $e_P \in [0, e_Z)$.

If $e_P < e_Z$, then as wealth increases, it will reach the point where $U'(p)$ has declined to equate $\theta U'(p) = (1 - \theta)b$, which we have defined as $\tilde{p}$. At that point, $m^*(W) = W - f - \tilde{p}$.

Then

$$m^*(W) = \begin{cases} 0 & W + \mathcal{D}(\tilde{f}, B(0)) \in [0, e_P) \\ W + \mathcal{D}(\tilde{f}, B(0)) - e_P & W + \mathcal{D}(\tilde{f}, B(0)) \in [e_P, e_Z] \end{cases} \quad (79)$$

B. Case B: If $e_P > e_Z$, then as wealth increases, it will pass through e_Z before e_P . So, no wealth be allocated to private benefits for the entirety

of the domain $[0, e_Z]$. Then

$$m^*(W) = \begin{cases} 0 & W + \mathcal{D}(\tilde{f}, B(0)) \in [0, e_Z] \end{cases} \quad (80)$$

- ii. **Domain** $W + \mathcal{D}(\tilde{f}, B(0)) \geq e_Z$ Then $m^*(W) = 0$ as marginal benefit loads on $\mathbf{z}^*$.

Combining both cases yields the stated piecewise form of $m^*(W)$.

■